

Thickness-aware Silvaco ATLAS model of ultrathin W-doped In₂O₃ (IWO) transistors

Complete report: literature theory, experiment, model definition, deck annotation, calibration, results, limitations

Package: IWO_PHYSICS_CONSTRAINED_MODEL_V1 (built 2026-09-20/21). Simulator: Silvaco DeckBuild 5.0.10.R + ATLAS 5.28.1.R, local licence, 32-bit build. Author of the runs: this session (29 launches, results/RUN_INDEX.csv). Every simulated number in this report comes from one of those launches; every material number carries a provenance label from the fixed set MEASURED_TARGET_DEVICE, MEASURED_RELATED_MATERIAL, DFT_TARGET_IWO, DFT_PURE_IN2O3_PROXY, LITERATURE_IWO, LITERATURE_IN2O3_PROXY, CALCULATED, FITTED, NUMERICAL, ASSUMED. Where the supplied inputs cannot fix a quantity the report says NOT DETERMINED FROM AVAILABLE DATA.

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1. Purpose and scope

The goal was a single, source-traceable ATLAS description of the sputtered ~2 % W:In₂O₃ channel that (a) contains the thickness physics explicitly, through laws for the band edges, effective mass, density of states, sub-gap states and background donors, instead of a separate parameter set per device; (b) is run on the real device structure; (c) is compared with the measured transfer curves of the 2.0, 6.3, 13.2 and 31.8 nm devices using identical metric extraction for measurement and simulation; and (d) is documented so that a reader can see, for every line of the deck, where its value comes from and why it exists.

The final set, at the user's request, is four launches: the 2.0 nm and 13.2 nm devices (calibrated), the 6.3 nm device run with the same parameters as an out-of-sample validation, and, after that validation had shown a threshold miss, the 6.3 nm

device tuned with one device-specific flat-band offset and its own band mobility. The 31.8 nm device is excluded from the final set; the evidence gathered on it is reported in Section 12.

Two earlier calibrations of the same data existed (GPT "Astra 6" package of 2026-09-12 and the Claude V0 package of 2026-09-10/21). Both reached 0.02-0.05 decades of agreement but with four to six tuned parameters per device. They were used here as numerical baselines and as audit material (Section 11), not as sources of physics.

2. The experiment: device, measurement, what is and is not known

2.1 Device (Januar et al. 2026, Sec. 2.1; user's schematic)

Layer (bottom to top)	Value	Verified in	Status
Substrate	n+ Si	paper Sec. 2.1	screened by the continuous TiN gate; not in the electrical domain
Gate	TiN, 50 nm (PVD)	paper; schematic "G = TiN = 50 nm"	MEASURED_TARGET_DEVICE
Gate dielectric	HfO2 ~15 nm (ALD, 250 C) then Al2O3 ~2 nm; Al2O3 touches the channel	paper (growth rates); nominal thicknesses from the schematic	MEASURED_TARGET_DEVICE
Channel	sputtered IWO, "~2 % W", t = 2.0 / 6.3 / 13.2 / 31.8 nm (XRR)	paper (2-13 nm series); workbook columns	MEASURED_TARGET_DEVICE; the 31.8 nm film lies outside the published series
Source/drain	Pd, 70 nm, on top of the channel	paper; schematic "S/D = Pd = 70 nm"	MEASURED_TARGET_DEVICE
Anneal	RTA 150 C, 600 s	paper	process context
W / L	290 um / 20 um	schematic	MEASURED_TARGET_DEVICE
VDS during the transfer sweep	0.7 V	schematic	MEASURED_TARGET_DEVICE (figure)
Top of the channel	air (no cap)	none shown	no fictitious cap added

The composition "2 % W" is not defined in the source (atomic, cation or mol% WO3; target or film). It enters the model only as the identity of the material. It is never converted into a donor density: 2 at.% W corresponds to about 6e20 W atoms per cm3, while every calibration of these curves needs an electrically active donor density three to four orders of magnitude smaller.

2.2 Measurement (workbook IWO-TFT (1).xlsx, sha256 ce97eccd...d47d12, preserved unchanged in data/)

Sheet1, rows 2-122: columns A:B = 6.3 nm, C:D = 13.2 nm, E:F = 31.8 nm, G:H = 2.0 nm. Each curve has 121 points from -3 V to +3 V in 0.05 V steps, all currents positive, VG strictly increasing. The headers carry no units. The unit A/um (current per

micrometre of width) and $V_{DS} = 0.7$ V are taken from the schematic and the PowerPoint slide. The currents were divided by $W = 290$ μm exactly once when the clean CSV was produced (data/experimental_clean.csv); the simulation uses a 1 μm width so its currents are numerically in A/ μm . Nothing is divided by 290 a second time.

t (nm)	I_d min (A/ μm)	I_d at +3 V (A/ μm)	off-band median, -2..-0.5 V	remark
2.0	1.08e-16 (single point at -3 V)	5.09e-7	4.6e-15	floor $\sim 5\text{e-}15$
6.3	5.9e-15	4.03e-7	1.5e-13	floor $\sim 1.5\text{e-}13$
13.2	6.4e-12	3.07e-6	6.6e-12	floor $\sim 6.6\text{e-}12$
31.8	2.6e-7	3.59e-6	2.7e-7	always on; on-state saturates

NOT DETERMINED FROM AVAILABLE DATA: sweep direction, dwell time, temperature, instrument floor, gate and source currents, the origin of the off-state plateaus.

2.3 Metrics extracted from the measurement (same code as for the simulation, Appendix B)

t (nm)	V_{th} (constant current $1\text{e-}9$ A/ μm)	SS_{min} (mV/dec)	SS over $1\text{e-}10..1\text{e-}8$ A/ μm (mV/dec)	μ_{FE} (cm^2/Vs)	I_{on} (A/ μm)	I_{on}/I_{off}
2.0	0.66 V	84	270	12.1	5.09e-7	4.7e9
6.3	0.64 V	115	295	10.9	4.03e-7	6.9e7
13.2	0.08 V	131	160	50.2	3.07e-6	4.8e5
31.8	none (always on)	-	-	49.2	3.59e-6	14

Two facts drive the whole modelling problem. First, the threshold of the 6.3 nm film equals that of the 2 nm film although the 13.2 nm film is 0.58 V lower. Second, the apparent mobility jumps by a factor five between 6.3 and 13.2 nm.

3. Theory from the literature

3.1 Bulk In_2O_3 and the transport-neutrality level

In_2O_3 is a wide-gap (experimental $\sim 2.9\text{-}3.2$ eV) n-type oxide whose conduction-band minimum is a single, dispersive s-like band. Its charge-neutrality level lies about 0.4 eV ABOVE the bulk conduction-band edge (King 2008; Robertson 2011; used by Si 2021, Lin 2022, Wang 2022). Consequently surfaces and defects tend to pin the Fermi level inside the conduction band, the material is naturally degenerate n-type, and contacts to metals tend to be low-barrier. This is why the ideal-Ohmic source/drain assumption is defensible and why the 0.4 eV value is used as a consistency check (Section 4.2).

Measured bulk values used as anchors (Stokey 2021, single crystal, optical Hall and ellipsometry): electron effective mass 0.208 m_0 at $n = 2.8\text{e}17$ cm^{-3} (zero-density extrapolation 0.18 m_0 , Feneberg), static permittivity 10.55, high-frequency permittivity 4.05. Evaporated films give $\epsilon_{ps} \sim 8.9$ (Hamberg 1986). The target process gives, by C-V on the same stack (Januar 2026 SI Fig. S1, values recorded from the online SI): IWO 9.30, In_2O_3 9.03, HfO_2 19.57.

3.2 Quantum confinement in nanometre-thin In2O3 (DFT, PBE)

Two first-principles studies give slab-versus-bulk band-edge shifts for pure In2O3:

source	slab	quantity	value
Lin 2022 (ACS Nano), H-passivated slabs in vacuum, ONCV/PBE	1.98 nm	Eg_PBE 1.27 eV vs bulk 0.94 eV	dEg = +0.33 eV
Lin 2022	0.95 nm	Eg_PBE 1.88 eV	dEg = +0.94 eV
Lin 2022	bulk / 3.52 / 1.98 / 0.95 nm	m* (PBE) 0.17 / 0.19 / 0.23 / 0.30 m0	dm* = +0.02 / +0.06 / +0.13 m0
Si 2021 (Nano Lett.), In2O3 on Al2O3, PAW/PBE	1.5 nm	Ec shift with Ev "almost unchanged"	dEc = +0.6 eV

PBE underestimates the absolute gap (0.94 eV vs ~3 eV experimentally), so absolute PBE gaps are never used. Only slab-minus-bulk differences are used, and only for pure In2O3: no DFT of W-doped slabs exists in the supplied material, and none could be executed here (no Quantum ESPRESSO on this machine; the inputs are prepared in `qe_workflow/`). W raises m* and flattens the conduction band (Januar 2026, qualitative), so the true IWO shifts are probably smaller; the uncertainty is carried as +/-50 %.

An infinite square well gives $dE \propto t^{-2}$. The fitted exponent (Section 4.1) is 1.38, smaller because of the finite barrier and the nonparabolic band.

3.3 Sub-gap density of states in amorphous oxide semiconductors

Amorphous oxide TFTs are described with a continuous sub-gap DOS: an exponential acceptor-like tail below the conduction band $g_{TA}(E) = N_{TA} \exp[(E - E_c)/W_{TA}]$, a deep acceptor Gaussian, an exponential donor-like valence tail, a donor Gaussian, and interface states D_{it} at the semiconductor/dielectric boundary. ATLAS implements exactly this family (DEFECTS statement). Wang 2022 measured for ALD In2O3 on HfO2: $D_{it} \sim 6.3e11 \text{ cm}^{-2} \text{ eV}^{-1}$ and bulk sub-gap DOS $3.3e20 \text{ cm}^{-3} \text{ eV}^{-1}$ at ND $\sim 1e20$; Lin 2022 inferred $D_{it} \sim 6e11$ from a 63.5 mV/dec swing; Fan 2021 (sputtered IWO, a different device) used Eg 3.05 eV, χ 4.30 eV and a deep-acceptor amplitude of $2.5\text{--}3.7e16 \text{ cm}^{-3} \text{ eV}^{-1}$. These are proxies (pure In2O3 on HfO2, or a different IWO process) and are labelled as such.

3.4 The analytic framework of the target paper (Januar 2026)

The paper describes the transfer curve by a blended current

$$I_D = I_{blend} + I_{off}, \text{ with } I_{blend} = [(1/I_{Dc})^m + (1/I_{Dt})^m + (1/I_{Dit})^m]^{(-1/m)} \quad (1a, 1b)$$

$$I_{Dc} = (W/L) \Gamma_c V_{ov}^2 \text{ (band-like conduction through extended states)} \quad (1c) \quad I_{Dt} = (W/L) \Gamma_t V_{ov}^\gamma \text{ (percolative conduction through tail states)} \quad (1d) \quad I_{Dit} = I_{off} \exp[(V_G - V_{FB})/(SS/\ln 10)] \text{ (interface-trap-mediated diffusion)} \quad (1e) \quad I_{off} = (W/L) q N_t \tau_s n_{FB} [\exp(-qV_D/kT) - 1] \quad (1f)$$

with $V_{ov} = (1/\kappa) \ln[1 + \exp(\kappa(V_G - V_{FB}))]$. The tail-channel strength is converted to a volumetric tail density by

$$N_t = \text{Cox}^2 / (2 \theta_t \epsilon_s k T_t) \cdot [\mu_c N_c \epsilon_s k T / (\Gamma_t \text{Cox} (\gamma - 1))]^{(2/\gamma)} \quad (3)$$

and the sub-threshold occupation by $n_t \approx \theta_t N_t \exp[(E_F - E_c)/(k T_t)]$ (8). T_t is the characteristic tail temperature ($k T_t = W_{TA}$ in ATLAS language). The paper reports, for the thickness series, that N_t rises by about a factor four from 13.2 to 2 nm while the peak mobility falls from ~27.4 to ~5.1 cm²/Vs; for the bias-stress devices it reports $N_t(2 \text{ nm}) = 5.3e19 \text{ cm}^{-3}$ and $N_t(10 \text{ nm}) = 3.39e18 \text{ cm}^{-3}$ (different device set, θ_t convention).

Mobility is modelled by the carrier-partition relation

$\mu_{\text{eff}}(\text{VG}) = n_{\text{free}} / (n_{\text{free}} + n_{\text{tail}} + n_{\text{it}}) \cdot \mu_{\text{max}} \text{ (5)}$

and the surface-roughness law

$\mu_{\text{sr}}(\text{Vov}, t) = \mu_0 / (1 + \theta_r \text{ Vov}) \cdot (1 - \Delta_{\text{sr}}/t)^2 \text{ (6)}$

whose first factor is the Ando surface-roughness roll-off and whose second factor is the quadratic collapse when the film thickness approaches the effective roughness Δ_{sr} ; the paper extracts $\Delta_{\text{sr}} \approx 2.87 \text{ \AA}$ for IWO ($3.75 \text{ \AA}$ for In_2O_3). Off-state current is stated to scale as $I_{\text{off}} \propto t_{\text{s}} n_{\text{FB}} \text{ (4)}$.

How each element maps into ATLAS is given in Section 4: Equation (5) is not inserted, it is produced self-consistently by the DOS occupancy of the solver; the roughness factor of Eq. (6) multiplies the band mobility; the Ando roll-off (θ_r) is NOT insertable natively (Section 12); Eq. (4) is used only as a consistency check.

3.5 Thickness trends from other IWO work

Kim 2024 (RF-sputtered 1 wt% $\text{WO}_3\text{:In}_2\text{O}_3$ on SiO_2 , different process) finds bulk/border trap densities of 2.4, 3.1 and $14.6 \times 10^{18} \text{ cm}^{-3} \text{ eV}^{-1}$ at 10, 20 and 30 nm, i.e. a strong rise for thick films. This supports a background-donor/defect density that rises for the 31.8 nm film (Section 4.6), but it is a different process and is labelled LITERATURE_IWO (different process).

3.6 Contacts

In_2O_3 contacts are low-barrier because of the CNL position (Section 3.1); scaled ALD In_2O_3 devices reach contact resistances below 0.1 ohm-mm (Si 2022). No transfer-length data exist for the target Pd/IWO contacts, so the contacts are ASSUMED Ohmic; for the 31.8 nm film the measured total resistance at 3 V (672 ohm) bounds any series resistance.

4. From the theory to one material model M(t)

The material model is config/iwo_material_model.yaml; the laws are fitted and evaluated by scripts/material_laws.py (evidence table tables/THICKNESS_LAW_EVIDENCE.csv, evaluated laws tables/THICKNESS_LAWS.csv). The deck generator scripts/build_iwo_decks.py evaluates M at the requested thickness and writes the deck with a provenance header. Category labels used in the decks: A intrinsic, B confinement, D bulk defects, E interface, F transport, G contacts, H numerical.

4.1 Band gap and electron affinity: Eg(t), chi(t) [A + B]

Definition: $E_g(t) = E_{g_bulk} + dE_{g_QC}(t)$; $\chi(t) = \chi_{_bulk} - dE_c(t)$; $dE_c = dE_g$ (the whole confinement shift assigned to the conduction band, E_v fixed, as found by Si 2021).

Law: $dE_{g_QC}(t) = 0.921 \cdot t^{-1.382} \text{ eV}$ (t in nm), a log-log least-squares fit through the three slab-minus-bulk PBE points (+0.33 eV at 1.98 nm, +0.94 eV at 0.95 nm, +0.60 eV at 1.5 nm).

Bulk anchors: $E_{g_bulk} = 3.05 \text{ eV}$, $\chi_{_bulk} = 4.30 \text{ eV}$ (Fan 2021, sputtered IWO of a different device; LITERATURE_IWO). The absolute χ is uncertain by $\sim 0.3 \text{ eV}$ and is degenerate with the gate work function and any fixed charge; only thickness DIFFERENCES of χ are used as physics. The absolute alignment is absorbed by one shared fitted quantity (Section 4.8).

Values:

t (nm)	dEg_QC (eV)	Eg = EG300 (eV)	chi = AFFINITY (eV)
2.0	0.353	3.403	3.947
6.3	0.072	3.122	4.228

t (nm)	dEg_QC (eV)	Eg = EG300 (eV)	chi = AFFINITY (eV)
13.2	0.026	3.076	4.274
31.8	0.008	3.058	4.292

Why it exists: it is the only mechanism in the inputs that shifts the 2 nm turn-on positive relative to the 13.2 nm film by a physically-derived amount (0.35 eV of affinity, i.e. ~ 0.35 V at fixed gate work function). Approach A: because the shift is in Eg/chi/ m^* /Nc explicitly, no BQP or Schrodinger quantum correction is enabled in ATLAS (that would double count).

Label: DFT_PURE_IN2O3_PROXY for the shift, LITERATURE_IWO for the anchors, CALCULATED for the evaluated values. TNL consistency check: with the transport-neutrality level fixed in absolute energy 0.4 eV above bulk Ec, TNL - Ec(t) = 0.4 - dEc(t) = +0.05 eV at 2 nm; the Fermi level is expected to sit lower in the gap for thinner films and the threshold to be more positive, which is what the 2 vs 13.2 nm data show.

4.2 Effective mass and conduction-band density of states: $m^*(t)$, Nc(t) [A + B]

Law: $m^*(t) = 0.208 + 0.131 \cdot t^{-1.412} m_0$. The bulk value 0.208 m_0 is measured (Stokey 2021, MEASURED_RELATED_MATERIAL); the increment is the PBE slab-minus-bulk increase of Lin 2022 (+0.02, +0.06, +0.13 m_0 at 3.52, 1.98, 0.95 nm), a proxy.

$N_c(t) = 2 (2\pi m k T / h^2)^{3/2} = 2.509e19 (m/m_0)^{3/2} \text{ cm}^{-3}$ at 300 K.

t (nm)	$m^* (m_0)$	Nc = NC300 (cm-3)
2.0	0.257	3.27e18
6.3	0.218	2.55e18
13.2	0.211	2.44e18
31.8	0.209	2.40e18

Why: Nc sets the relation between the Fermi level and the free-electron density (Eq. 2 of the paper); both earlier calibrations used Nc = 5e18, which corresponds to $m^* = 0.34 m_0$ and has no source. With the measured mass the DOS is lower, which the shared electrostatic value compensates. Nv = 1e19 cm-3 is ASSUMED; holes are not solved (Section 7).

4.3 Permittivity [A]

eps_IWO = 9.30, from C-V on the target process (Januar 2026 SI Fig. S1; MEASURED_TARGET_DEVICE), shared over thickness. The bulk-crystal 10.55 and the film 8.9 are kept as sensitivity bounds; in a 2 nm film in series with the 3.86 nm EOT gate stack the semiconductor capacitance term changes the accumulation charge by well under 1 %.

4.4 Conduction-band tail states: Nt(t), W_TA [D]

ATLAS tail: $g_{TA}(E) = N_{TA} \exp[(E - E_c)/W_{TA}]$, acceptor-like (negative when filled). Its integral over the gap is $N_t \approx N_{TA} \cdot W_{TA}$. $W_{TA} = k \cdot T_t$.

Law: $N_t(t) = 2.0e19 \cdot (2 \text{ nm} / t)^{0.75} \text{ cm}^{-3}$, $W_{TA} = 0.040 \text{ eV}$ shared ($T_t = 464 \text{ K}$). Anchors: the real-ATLAS calibrations of both earlier lineages give $N_{TA} \cdot W_{TA} = 2.0\text{-}2.2e19 \text{ cm}^{-3}$ at 2 nm, 1.3-1.5e19 at 6.3 nm and 2.5-4.6e18 at 13.2 nm (exponent 0.7-0.8), and the paper reports $\sim 4\times$ between 13.2 and 2 nm (exponent 0.73); the paper's 5.3e19 at 2 nm uses the θt

convention of Eq. (8) and is of the same order. Absolute T_t is not printed in the paper (only ΔT_t), so W_{TA} is FITTED once and shared.

t (nm)	Nt (cm-3)	N_TA = Nt/W_TA (cm-3 eV-1)
2.0	2.00e19	5.00e20
6.3	8.46e18	2.11e20
13.2	4.86e18	1.21e20
31.8	2.51e18	6.28e19

Why: the tail sets the subthreshold slope and the free/trapped partition of Eq. (5). Because the solver fills these states according to Fermi statistics at every bias, the carrier-partition mobility of the paper emerges from the DOS without a second empirical factor.

Deep acceptor Gaussian: $N_{GA} = 5e16 \text{ cm}^{-3} \text{ eV}^{-1}$ at $E_c - 0.6 \text{ eV}$, width 0.15 eV , shared, ASSUMED (Fan 2021 range); the DC transfer curves are insensitive to it. Valence tail $N_{TD} = 0$ and bulk donor Gaussian $N_{GD} = 0$ (ASSUMED; the effective background donors are represented by N_{d_eff} instead). Capture cross sections $1e-15 \text{ cm}^2$ for all states (ASSUMED; not identifiable from DC curves).

4.5 Interface states D_{it} [E]

Acceptor-like Gaussian sheet of peak $3e11 \text{ cm}^{-2} \text{ eV}^{-1}$ centred 0.3 eV below E_c , width 0.12 eV , shared over thickness, ASSUMED: literature $\text{HfO}_2/\text{In}_2\text{O}_3$ interfaces give $\sim 6e11$ (Lin 2022, Wang 2022) and the target interface is $\text{IWO}/\text{Al}_2\text{O}_3$, which Januar reports as smoother. Implementation: a 0.25 nm IWO layer next to the Al_2O_3 (region 2) carries a volume Gaussian of amplitude $3e11 / 2.5e-8 \text{ cm} = 1.2e19 \text{ cm}^{-3} \text{ eV}^{-1}$, moment-matched with the bulk deep Gaussian (the printed values $1.16757e19$ at 0.3016 eV , width 0.1240 eV are the moment-matched combination of the sheet with the $5e16$ bulk Gaussian). Halving the layer thickness changes I_d by 0.009 decades (Section 10).

D_{it} is kept separate from the fixed charge Q_f : D_{it} changes the swing and the free/trapped partition; Q_f shifts the flat band only.

4.6 Effective background donor density $N_{d_eff}(t)$ [D]

Law: $N_{d_eff}(t) = 2.5e17 \cdot [1 + (t / 20 \text{ nm})^4] \text{ cm}^{-3} \rightarrow 2.50e17, 2.52e17, 2.97e17, 1.85e18$ for $2.0, 6.3, 13.2, 31.8 \text{ nm}$. FITTED law (three coefficients for four films): both earlier lineages needed $\sim 2-3e17$ for the thin films and $>1e18$ for the 31.8 nm film; Kim 2024's rising trap density for thick films supports the direction. This is the electrically active density, not the W concentration (Section 2.1). The flat-band free-electron density n_{FB} is not an input; it is the solver's equilibrium output (probe `avg_n`, exported per run).

4.7 Mobility [F]

$\mu_0(t) = \mu_{band}(t) \cdot (1 - \Delta sr/t)^2$ with $\Delta sr = 0.287 \text{ nm}$ (Januar Fig. 5f; the prose gives $2.87 \text{ \AA}$; the figure axis unit is ambiguous and this is noted). Roughness factor: $0.734, 0.911, 0.957, 0.982$.

μ_{band} is FITTED per thickness: 18.13 (2.0 nm , stage 6), 11.69 (6.3 nm , stage 8; the validation run used the earlier value 12.4 unchanged), 61.9 (13.2 nm , stage 6). The step between 6.3 and 13.2 nm has no defensible law in the inputs (hypothesis: percolation/crystallinity; Januar Figs. S2-S4 show nanoscale grains; Kim 2024 finds an optimum near 20 nm).

The deck value $MUN = \mu_0(t)$: $13.30, 11.30, 59.24 \text{ cm}^2/\text{Vs}$. The gate-voltage dependence of the effective mobility comes from the DOS occupancy (Eq. 5 realised by the solver); the Ando roll-off $\mu_0/(1 + \theta_r V_{ov})$ is NOT inserted: this ATLAS version

ignores MOBILITY statements between SOLVE statements (verified in V0 run_0008) and its field-dependent surface models (CVT, Lombardi) are silicon calibrations with no IWO basis. Hole mobility 0.01 (irrelevant, holes not solved).

4.8 Fixed interface charge Q_f and gate work function [E, G]

Gate work function 4.70 eV, ASSUMED (TiN literature 4.5-4.9 eV), held fixed. Q_f at the IWO/ Al_2O_3 interface: ONE value shared by all thicknesses, FITTED in stage 3: $Q_f = +1.73\text{e}12 \text{ cm}^{-2}$, equivalent to a flat-band shift $dV_{fb} = -q Q_f / C_{ox} = -0.31 \text{ V}$. It is classified as an effective electrostatic correction: it absorbs the unmeasured absolute χ_{bulk} , the unmeasured TiN work function and any true fixed charge, which one transfer curve per device cannot separate. It is not used to hide a wrong affinity: with $Q_f = 0$ the two calibrated films were offset by the SAME +0.3 V (Section 8), which is exactly what a common alignment error looks like and what a thickness-dependent χ error would not.

4.9 Contacts and series resistance [G]

Source/drain: Ohmic (no CONTACT work function), ASSUMED (Section 3.6). Series resistance zero for the thin films; a $9.2\text{e}4 \text{ ohm}\cdot\mu\text{m}$ lumped value (ATLAS CONTACT RESISTANCE is in $\text{ohm}\cdot\mu\text{m}$ in 2-D) was a FITTED hypothesis for the 31.8 nm film only and is not part of the final set.

4.10 C_{ox} and the gate stack

$C_{ox} = \epsilon_0 / (15 \text{ nm} / 19.57 + 2 \text{ nm} / 9.0) = 8.955\text{e}-7 \text{ F/cm}^2$ (EOT 3.86 nm), $C_{ox}/q = 5.59\text{e}12 \text{ cm}^{-2} \text{ V}^{-1}$. HfO₂ 19.57 is MEASURED_TARGET_DEVICE (SI Fig. S1); Al_2O_3 9.0 is ASSUMED (not reported). Both are set explicitly in the deck because the ATLAS built-in permittivities of the named materials are not the measured values.

5. The device as represented in ATLAS

Two representations are generated by the same script and were both run:

A. Reduced electrical deck (calibration stages, runs 0001-0009): IWO regions 1+2 (region 2 is the 0.25 nm interface-regularization layer), oxide regions 3 (Al_2O_3) and 4 (HfO₂) with the permittivities set by region, gate = Dirichlet boundary at the bottom of the HfO₂ with the TiN work function, source/drain = Dirichlet boundaries on the IWO top surface over 0-2 μm and 22-24 μm . In DC a continuous metal is an equipotential, so replacing its volume by a boundary with the same work function changes neither the field in the dielectric nor the contact condition and removes nodes (the 32-bit ATLAS build is limited to ~30k nodes).

B. Physical structure (final runs 0012-0014, at the user's request): the same channel and dielectric regions with material= Al_2O_3 and material=HfO₂, a 50 nm Conductor volume for the TiN gate carrying the gate electrode (TiN is not in the ATLAS metal table; the work function 4.70 eV is set on CONTACT), a 70 nm Air region above the channel, and 70 nm x 2 μm Palladium electrode volumes for source and drain on the IWO top surface with no imposed work function.

Electrical equivalence of A and B was verified in the solver with identical parameters: drain currents agree to within 0.1 % at every compared bias (run_0012 vs run_0008 at 2 nm; run_0013 vs run_0009 at 13.2 nm; Section 10).

Coordinates (μm): x = 0..24 (source 0-2, channel 2-22, drain 22-24); y = -t-0.07..-t Air/Pd, -t..0 IWO, 0..0.002 Al_2O_3 , 0.002..0.017 HfO₂, 0.017..0.067 TiN volume. The 2 μm contact overlap is ASSUMED (not reported); with Ohmic top contacts and a 20 μm channel the current is insensitive to it. MESH WIDTH = 1 μm .

6. The final deck, line by line

The 2.0 nm deck of run_0012 (results/runs/run_0012_2p0_v2_full_stepped_final/device.in, also decks/iwo_2p0nm.in) is annotated below. The 13.2 nm and 6.3 nm decks differ only where stated in 6.9.

6.1 Provenance header (lines 1-31)

Comment lines only. They print, for every material quantity, the value written below, its category letter, its provenance label and its source or law, e.g.

```
# [B] dEg_QC(t) = 0.3533 eV | DFT_PURE_IN2O3_PROXY | a*t^-p, a=0.9205, p=1.3815 (Lin2022 slabs, Si2021)
```

They exist so that the deck alone is auditable without the YAML.

6.2 Mesh (lines 32-52)

go atlas starts the device simulator inside DeckBuild.

mesh width=1.0 sets the third dimension to 1 um, so every terminal current is numerically A/um.

x.mesh loc=0 spac=0.3 ... x.mesh loc=24 spac=0.3: eleven lateral mesh lines. Spacing is 0.3 um under the contacts, refined to 0.05 and 0.01 um at the contact edges (x = 1.8-2.2 and 21.8-22.2 um) where the current enters the channel, and 0.7 um in the middle of the 20 um channel where nothing varies laterally. Halving all lateral spacings changed the current by < 0.005 decades (Section 10, check A).

y.mesh loc=-0.072 spac=0.02: top of the Air/Pd layer (70 nm above the film), 20 nm spacing; nothing electrical happens there.

y.mesh loc=-0.002 spac=0.00025: the IWO top surface at y = -2 nm; 0.25 nm spacing through the film (8 intervals across 2 nm; for thicker films the spacing is min(1 nm, t/8)).

y.mesh loc=-0.00025 spac=6.25e-05, loc=-0.000125, loc=0 spac=6.25e-05: the 0.25 nm interface layer is resolved with 0.0625 nm spacing (four intervals) so the regularized sheet DOS and the accumulation layer next to the Al2O3 are resolved.

y.mesh loc=0.002 spac=0.00025: Al2O3, 0.25 nm spacing. y.mesh loc=0.017 spac=0.001: HfO2, 1 nm spacing. y.mesh loc=0.067 spac=0.02: the gate metal volume, coarse.

6.3 Regions and electrodes (lines 53-65)

region num=1 user.material=IWO ... y.min=-0.002 y.max=-0.00025: the IWO bulk. user.material=IWO declares a user-defined material name whose parameters are all supplied by the MATERIAL statement below.

region num=2 user.material=IWO ... y.min=-0.00025 y.max=0: the same material; a separate region number is needed only so that the DEFECTS statement can give this 0.25 nm layer the additional interface Gaussian.

region num=3 material=Al2O3 ... y.min=0 y.max=0.002 and region num=4 material=HfO2 ... y.max=0.017: the two dielectrics as named ATLAS insulators (their permittivities are overridden below).

region num=6 material=Conductor ... y.min=0.017 y.max=0.067: the 50 nm TiN gate as a conductor volume.

region num=7 material=Air ... y.min=-0.072 y.max=-0.002: air above the channel (permittivity 1); the Pd volumes are carved out of it by the electrode statements.

electrode name=gate x.min=0 x.max=24 y.min=0.017 y.max=0.067: the gate electrode occupies the whole conductor volume.

electrode name=source material=Palladium x.min=0 x.max=2 y.min=-0.072 y.max=-0.002 and electrode name=drain material=Palladium x.min=22 x.max=24 ...: 70 nm x 2 um Pd volumes on top of the film ends. No CONTACT work function is set for them, so ATLAS treats them as Ohmic boundaries (ASSUMED; Section 3.6). This is verified by the equivalence with the reduced deck.

6.4 Doping (lines 66-67)

doping uniform n.type conc=2.50025e+17 region=1 (and region=2): the effective background donor density N_{d_eff} (2 nm) of Section 4.6. It is a fitted law value, not the W content.

6.5 Material (lines 68-78)

material material=IWO user.group=semiconductor user.default=silicon \: declares IWO as a semiconductor; user.default=silicon only tells the parser which parameter template to start from. Every parameter that matters at 300 K is overridden on the following continuation lines; nothing silicon-specific survives (no silicon mobility models are enabled, and all band, DOS, mobility and lifetime numbers are given explicitly).

eg300=3.40331: $E_g(2\text{ nm}) = 3.05 + 0.353$ (Section 4.1). affinity=3.94669: $\chi(2\text{ nm}) = 4.30 - 0.353$. permittivity=9.3: measured (Section 4.3).

nc300=3.27444e+18: N_c from $m^*(2\text{ nm}) = 0.257 m_0$ (Section 4.2). nv300=1e+19: assumed, unused in practice.

mun=13.3: $\mu_0(2\text{ nm}) = 18.13 \times 0.7336$ (Section 4.7). mup=0.01: hole mobility, irrelevant.

taun0=1e-06 taup0=1e-06: SRH lifetimes, ASSUMED; thermal generation is 13-17 decades below the measured floors, so they are not tuned.

material region=3 permittivity=9.0 and material region=4 permittivity=19.57: Al_2O_3 (assumed 9.0) and HfO_2 (measured 19.57). They exist because the built-in values for the named materials are not the measured ones; together they fix $C_{ox} = 8.955\text{e-}7\text{ F/cm}^2$.

contact name=gate workfunction=4.7: TiN work function, ASSUMED (Section 4.8).

6.6 Models and sub-gap DOS (lines 79-98)

models fermi srh temp=300 print: Fermi-Dirac statistics (the accumulation layer of a 2 nm film with $N_c \sim 3\text{e}18$ becomes degenerate; Boltzmann would overestimate n); SRH recombination as background only; 300 K; print echoes the parameters actually used into the log for audit.

defects region=1 continuous numa=96 numd=48 \: continuous sub-gap DOS for region 1, discretised with 96 acceptor and 48 donor energy levels (doubling them changes I_d by 0.006 dec, Section 10).

nta=5e+20 wta=0.04: conduction-band tail, $N_{TA} = N_t/W_{TA} = 2.0\text{e}19/0.040$ (Section 4.4), $W_{TA} = 0.040\text{ eV}$. ntd=0 wtd=0.1: no valence tail.

nga=5e+16 ega=0.6 wga=0.15: deep acceptor Gaussian, 0.6 eV below E_c (EGA is measured from E_c in this ATLAS version), assumed and insensitive.

ngd=0 egd=3.30331 wgd=0.1: no donor Gaussian; EGD is measured from E_v , so the placeholder energy is $E_g - 0.1$ (it has no effect at zero amplitude).

sigtae=1e-15 ... siggdh=1e-15: electron and hole capture cross sections of the tail and Gaussian states, all $1\text{e-}15\text{ cm}^2$, ASSUMED (not identifiable from DC curves).

afile=acceptor_r1.dat dfile=donor_r1.dat: ATLAS writes the DOS it actually used to these files, which are kept in every run directory as evidence.

defects region=2 ...: identical tail; nga=1.16757e+19 ega=0.301554 wga=0.123975 is the interface sheet regularized into the 0.25 nm layer (peak $3\text{e}11\text{ cm-}2\text{ eV-}1 / 0.25\text{ nm} = 1.2\text{e}19\text{ cm-}3\text{ eV-}1$, moment-matched with the $5\text{e}16$ bulk Gaussian, Section 4.5).

interface qf=1.73e+12 x.min=0 x.max=24 y.min=-0.0000001 y.max=0.0000001: the fixed charge on the IWO/ Al_2O_3 interface ($y = 0$), the single shared fitted electrostatic value (Section 4.8), $dV_{fb} = -0.31\text{ V}$.

6.7 Numerical method, output, probes and equilibrium (lines 99-108)

`method newton trap maxtraps=10 itlimit=80 climit=1e-6 ir.tol=1e-19 cr.toler=1e-17 xandrnorm carriers=1`
electrons: full Newton with up to 10 bias-step halvings on non-convergence (MAXTRAPS may not exceed 10), 80 iterations per bias, a tightened absolute current tolerance (IR.TOL) far below the measured floor, and XANDRNORM, which accepts a bias point only when BOTH the update norm and the residual norm are satisfied. Without it, in V0, ATLAS accepted deep-depletion points with a $1\text{e-}15\text{ A}/\mu\text{m}$ continuity residual (visible as $|I_d + I_s| \sim 3\text{e-}15\text{ A}$). `carriers=1 electrons` solves electrons only: hole currents were $1\text{e-}38..1\text{e-}55\text{ A}$ in every two-carrier run of both lineages and their relative-update norm was pure noise that blocked XANDRNORM convergence. `cr.toler` is set to $\max(1\text{e-}17\text{ A}, 1\text{e-}5 \times \text{smallest measured active current})$ and this strict criterion is used up to the measured threshold (0.7 V for 2 nm).

`output con.band val.band e.mobility:` requests the band edges and the electron mobility in the saved structure files.

`probe name=chan_mob x=12 y=-0.001 n.mob dir=0, probe name=chan_n x=12 y=-0.001 n.conc, probe name=avg_n region=1 average n.conc ...:` native solver quantities recorded at every bias: electron mobility and free-electron density at the channel centre, and the film-averaged free-electron density (whose $V_G = 0$ value is n_{FB}).

`solve init:` the equilibrium solution (all contacts at 0 V). `save outf=equilibrium.str:` saved for the equilibrium band diagram.

6.8 Bias ramp and stepped sweep (lines 109-131)

`solve vstep=-0.1 vfinal=-3 name=gate:` ramps the gate from 0 to -3 V in 0.1 V steps (deep depletion of the film), continuing from the equilibrium solution. `solve vstep=0.1 vfinal=0.7 name=drain:` ramps the drain to $V_{\text{DS}} = 0.7\text{ V}$.

`log outf=transfer.log:` opens the logged sweep.

`solve vgate=-3 vstep=0.2 vfinal=-1 name=gate:` the first logged block; VGATE gives the start (already the current bias), then 0.2 V steps in deep depletion where the current is at the solver floor and nothing is extracted.

`solve vstep=0.05 vfinal=0.7 name=gate:` 0.05 V steps, the measured grid, from -1 V to the measured threshold; threshold and swing are extracted here.

`save outf=near_vth.str:` structure at $V_G \approx V_{\text{th}}$ for the near-threshold band diagram.

`method ... cr.toler=5e-18 ^xandrnorm ...:` above threshold the default acceptance is restored (the on-state residual floor scales with the current; the strict criterion would never converge there).

`solve vstep=0.05 vfinal=1.5 name=gate` and `solve vstep=0.1 vfinal=3 name=gate:` the on-state, 0.05 V steps to 1.5 V (mobility peak region), 0.1 V steps to 3 V. In total 76 solved points; the measurement is compared at exactly these points.

`log off, save outf=final.str:` end of sweep, on-state structure.

`extract ... curve(v."gate", i."drain") outfile="idvg.dat"` (and gate, source currents, and the three probes): DeckBuild exports of the logged quantities. Gate and source currents are exported so that Kirchhoff's law $I_d + I_s + I_g = 0$ can be checked at every bias (Section 10).

6.9 Cut-line exports (lines 132-147)

`extract init infile="equilibrium.str" then curve(depth, impurity="Conduction Band Energy" material="All" mat.occno=1 x.val=12)` and the same for the valence band, electron concentration and electron quasi-Fermi level, at $x = 12\text{ }\mu\text{m}$ (channel centre), repeated for the near-threshold and on-state structures. These produce the band diagrams and electron-density profiles in plots/.

6.10 What differs in the 13.2 nm and 6.3 nm decks

line	2.0 nm	6.3 nm	13.2 nm
y.mesh through the film	0.25 nm	0.7875 nm (t/8)	1 nm
doping conc	2.50e17	2.52e17	2.97e17
eg300 / affinity	3.4033 / 3.9467	3.1224 / 4.2276	3.0761 / 4.2739
nc300	3.27e18	2.55e18	2.44e18
mun	13.30	11.30	59.24
nta (wta 0.04 everywhere)	5.00e20	2.11e20	1.21e20
interface qf	1.73e12	1.73e12	1.73e12
strict cr.toler / strict up to	1e-17 A / 0.7 V	2.28e-17 A / 0.65 V	7.88e-16 A / 0.1 V

Everything else (region 2 Gaussian, Dit, deep Gaussian, capture cross sections, contacts, models, mesh in x and in the dielectrics, sweep) is identical.

7. Solver and numerical choices, with reasons

choice	reason
Direct ATLAS device construction, no ATHENA	a process simulator cannot predict sputtered IWO chemistry, W incorporation or the amorphous DOS; ATHENA would only reproduce the geometry, which is known.
Drift-diffusion, classical	the confinement physics is carried explicitly by $E_g(t)$, $\chi(t)$, $m^*(t)$, $N_c(t)$ (Approach A); enabling BQP/Schrodinger on top would double count.
Fermi statistics	degenerate accumulation in a low- N_c film.
Electrons only	holes are numerically zero; including them destabilised strict convergence.
SRH background only	not tunable against the data; thermal generation cannot produce the measured off floors.
Strict acceptance below V_{th} , default above	see 6.7; verified in V0 runs 0006/0009/0010/0011.
96/48 DOS levels	x2 changes I_d by 0.006 dec.

choice	reason
0.25 nm interface regularization layer	halving it changes I_d by 0.009 dec.
Stepped sweep 0.2/0.05/0.1 V	verified equal to 0.05 V point-by-point solves (Section 10).
Native zeros kept	below $\sim 1e-17$ A/ μ m the solver returns 0 or $\pm 1e-19$ A/ μ m; these are counted, excluded from log residuals and never replaced by a floor.
32-bit build	meshes are kept below $\sim 30k$ nodes (the 13.2 nm x 0.5 mesh with 40k nodes crashes).

8. Calibration procedure: stages, launches, what was free at each stage

All launches are real, sequential, and logged (Appendix A). The scoring region "active" is the set of measured points above five times the measured off-band median, i.e. where the measurement carries device information; the score is the RMS of $\log_{10}(I_{sim}/I_{meas})$ over that region in decades.

stage	free quantities	launches	outcome
1-2. Laws only (Sections 4.1-4.7), $Q_f = 0$, WF 4.70, μ_{band} from the earlier calibrations	none tuned	run_0001 (2 nm), run_0002 (13.2 nm)	both curves have the measured SHAPE (SS_min sim/meas 81/84 at 2 nm, 146/131 at 13.2 nm) but sit +0.33 V (2 nm) and +0.29 V (13.2 nm) too positive; I_{on} -27 % / -21 %, mostly the lost overdrive; active RMSE 3.93 / 1.03 dec
3. One shared flat-band offset: $Q_f = +1.73e12$ cm ⁻² (dVfb -0.31 V)	1 shared	run_0003 (2 nm), run_0004 (13.2 nm)	2 nm 0.056 dec, dVth +0.02 V, I_{on} -7.3 %; 13.2 nm 0.086 dec, dVth -0.02 V, I_{on} -7.0 %. One number fixes both because the baseline offsets agreed to 40 mV
5. Cross-thickness test at 6.3 nm with the stage-3 parameters	0	run_0005	Vth 0.35 V vs 0.64 V measured (-0.29 V); shape reproduced (SS_min 108 vs 115); 0.68 dec
5b. Evidence run: one device-specific offset for 6.3 nm (Q_f 8.7e10)	1 device-specific	run_0007	0.082 dec, I_{on} +6 %. Not part of the shared model
5. 31.8 nm with laws + shared Q_f + series resistance	0 (+1 inherited)	run_0006	I_{on} -6 % but the bulk depletes below -1 V while the measurement is flat: always-on baseline not reproduced
6. Band mobility refit to I_{on} at fixed electrostatics	1 per film (declared FITTED)	run_0008 (2 nm, 16.8→18.13), run_0009 (13.2 nm, 57.6→61.9)	0.037 dec / 0.081 dec, I_{on} within 0.02 %

stage	free quantities	launches	outcome
aborted	31.8 nm back-surface hypothesis (run_0010, stopped in SOLVE INIT without error text); 6.3 nm mobility stage (run_0011, chain stopped by the user)		no currents produced
7. Final form: physical structure + stepped sweep, same parameters as stage 6	0 new	run_0012 (2 nm), run_0013 (13.2 nm), run_0014 (6.3 nm validation, shared parameters)	Section 9
8. 6.3 nm tuned (user request after the validation miss): the stage-5b device-specific offset ($Q_f 8.7e10$) plus the mobility stage ($\mu_{band} 12.4 \rightarrow 11.69$, removing the +6.1 % I_{on} of run_0007)	1 device-specific + 1 fitted mobility	run_0015	0.063 dec, I_{on} -0.01 %, V_{th} 0.65 vs 0.64 V, SS_{min} 111 vs 115 mV/dec

Free-parameter count for the final model: one shared electrostatic value (Q_f), one band mobility per film, and one device-specific flat-band offset for the 6.3 nm film; all laws untuned. Astra: per-device NTA, WTA, ND, μ plus a shared work function (13 numbers for three films). Claude V0: per-device NTA, WTA, ND, μ , dV_{fb} , Dit (22+ numbers for four films).

The sensitivities that these stage pairs measure directly are in `tables/SENSITIVITY_RESULTS.csv`: $dV_{th} = -0.309$ V for $dQ_f = 1.73e12$ cm⁻² (the solver reproduces $-q dQ_f/Cox = -0.3095$ V exactly); I_{on} scales linearly with μ_{band} (+7.9 % $\rightarrow$ +7.9 %); the 6.3 nm offset of $-1.64e12$ cm⁻² moves V_{th} by +0.294 V; changing the structure and sweep form changes nothing (< 0.2 % in any metric).

8.1 Counterfactual and one-at-a-time sensitivity runs (2026-09-22, runs 0016-0027)

No-confinement counterfactual ($dE_g = 0$, $m^* = \text{bulk}$, everything else at the final values):

film	reference	counterfactual	dV_{th}	dSS_{min}	dI_{on}	active RMSE
2.0 nm	run_0012	run_0016	-0.316 V	+11 mV/dec	+21 %	0.037 $\rightarrow$ 0.846 dec
13.2 nm	run_0013	run_0017	-0.023 V	-13 mV/dec	+1 %	0.081 $\rightarrow$ 0.114 dec

Reading: the confinement law moves the 2 nm film by 0.32 V and the 13.2 nm film by 0.02 V. Without it, fitting both thresholds would require fixed charges differing by $(0.316 - 0.023) \times Cox/q = 1.6e12$ cm⁻², i.e. a per-device electrostatic parameter. With it, one shared value fits both. This is the test that separates "the law is consistent with the data" from "the law is doing the work": it is doing the work.

One-at-a-time perturbations at 2 nm (each against run_0012; the changed input is the only difference):

case	change	dV_{th} (V)	dSS_{min} (mV/dec)	$d\mu_{FE}$	dI_{on}	RMSE (dec)
chi_bulk -0.1 eV	run_0020	+0.10	0	-0.7 %	-6.9 %	0.46
chi_bulk +0.1 eV	run_0021	-0.10	0	+0.6 %	+7.0 %	0.36

case	change	dVth (V)	dSS_min (mV/dec)	dmu_FE	dIon	RMSE (dec)
Nd0 x2	run_0022	-0.01	+1.6	+0.1 %	+0.7 %	0.054
Nt2 x1.5	run_0023	+0.08	+3.1	-6.8 %	-25 %	0.26
WTA +5 meV	run_0024	+0.02	+12.4	+0.3 %	+0.4 %	0.063
Dit x3	run_0025	+0.03	+9.4	-0.2 %	-1.8 %	0.098
eps_IWO 9.3 → 10.55	run_0026	0.00	-0.1	+0.6 %	+0.7 %	0.037
contact overlap 2 → 4 um	run_0027	0.00	0	-1.8 %	-2.2 %	0.038

What this establishes about identifiability at 2 nm: the threshold is controlled by chi (or equivalently Qf, WF) and, weakly, by the tail density; the swing by WTA and Dit, which are therefore separable from the electrostatic parameters; the on-current by the tail density through the free/trapped partition (x1.5 in Nt costs a quarter of Ion at unchanged mobility) and by chi through the overdrive. Nd, eps and the contact overlap are irrelevant for the 2 nm film, which confirms that the fitted electrostatics are not hiding a donor or geometry error. Not run: gate work function (identical action to Qf), m* alone, and any case at 6.3 nm.

9. Results

9.1 Final table (identical extraction for both curves; `tables/EXPERIMENTAL_VS_SIMULATION.md`)

t (nm)	run	role	Vth_cc exp / sim (V)	SS_min exp / sim (mV/dec)	SS 1e-10..1e-8 exp / sim	mu_FE exp / sim (cm ² /Vs)	Ion exp / sim (A/um)	Ioff exp / sim	active log RMSE	native zeros
2.0	run_0012	calibrated	0.66 / 0.68	84 / 73	270 / 294	12.1 / 11.3	5.09e-7 / 5.09e-7 (+0.0 %)	1.1e-16 / 5e-21 (min positive)	0.037 dec	24
13.2	run_0013	calibrated	0.08 / 0.05	131 / 151	160 / 195	50.2 / 49.0	3.07e-6 / 3.07e-6 (-0.0 %)	6.4e-12 / 3e-20	0.081 dec	37
6.3	run_0014	VALIDATION, nothing tuned	0.64 / 0.35	115 / 108	295 / 285	10.9 / 9.1	4.03e-7 / 5.11e-7 (+26.8 %)	5.9e-15 / 1e-21	0.678 dec	23
6.3	run_0015	TUNED: device-specific Qf 8.7e10 (+0.29 V) and μ _{band} 11.69	0.64 / 0.65	115 / 111	295 / 289	10.9 / 8.4	4.03e-7 / 4.03e-7 (-0.0 %)	5.9e-15 / 7e-22	0.063 dec	25

Ion/Ioff: measured 4.7e9, 6.9e7, 4.8e5; simulated values are lower bounds (> 1e14) because the simulated off-state is a native zero; the measured floors are not modelled and no percentage is reported.

9.2 Reading the residuals (plots//residual.png)

2.0 nm: the whole active region lies within ± 0.08 dec; a shallow -0.08 dec dip around $V_G \approx 1$ V (mobility peak region) and a slight positive residual right at turn-on. Simulated $SS_{\min}$ is 11 mV/dec steeper than measured.

13.2 nm: $+0.45$ dec at the single point where the measured curve leaves its 6.6×10^{-12} A/ μ m floor (the simulation has no floor to leave), -0.1 dec around 0.5 V, then within ± 0.03 dec above 1 V. Simulated $SS_{\min}$ is 20 mV/dec softer than measured; the 1×10^{-10} .. 1×10^{-8} band swing is 195 vs 160 mV/dec. Ion matches to 0.02 %.

6.3 nm (validation): the simulated curve is the measured curve shifted 0.29 V to the left; the subthreshold and on-state shapes are right ($SS_{\min}$ within 7 %, band swing within 4 %, apparent mobility within 17 %), so the +27 % on-current is an overdrive error, not a mobility error.

6.3 nm (tuned, run_0015): with the device-specific offset (Qf 8.7×10^{10} cm $^{-2}$ instead of the shared 1.73×10^{12} , i.e. +0.29 V, equivalent to 1.6×10^{12} cm $^{-2}$ less positive interface charge) and the band mobility refitted to Ion (11.69 cm 2 /Vs), the film is reproduced as well as the other two: 0.063 dec, Ion within 0.02 %, Vth within 10 mV, $SS_{\min}$ within 4 mV/dec; the residual stays within ± 0.1 dec over the whole active region ($+0.2$ dec at the single point where the measured curve leaves its 1.5×10^{-13} A/ μ m floor, and a broad $+0.1$ dec hump around 1.5-2 V where the measured apparent mobility, 10.9, exceeds the simulated 8.4 cm 2 /Vs). The offset itself is the quantity the shared laws do not explain.

9.3 Physics conclusions supported by the runs

1. The pure-In $2O_3$ confinement proxy ($dE_c = 0.35$ eV between 2 and 13.2 nm), together with $N_c(t)$, $N_t(t)$ and $N_d(t)$, accounts for the measured 0.58 V threshold difference between the 2 and 13.2 nm films: the untuned laws misplaced both thresholds by the same +0.3 V, and one shared value corrected both. The counterfactual runs (Section 8.1) confirm that the law is doing the work: without it the two films would need fixed charges differing by 1.6×10^{12} cm $^{-2}$.
2. The subthreshold shapes of both calibrated films follow from the tail law with one shared W_{TA} and one shared D_{it} without tuning.
3. The band mobility is not predicted by anything in the inputs: 18.1 vs 61.9 cm 2 /Vs after the roughness factor.
4. The 6.3 nm film contradicts every monotonic thickness law: its threshold sits at the 2 nm value. Cause NOT DETERMINED FROM AVAILABLE DATA (candidates: sample-to-sample fixed charge or ageing, thickness or process deviation, non-monotonic donor/confinement behaviour between 2 and 6 nm). Both earlier calibrations had hidden this by tuning a per-device parameter.
5. The measured off-state plateaus are not thermal generation in this stack (13-17 decades margin) and cannot be attributed with the available data (no gate or source current records).

9.4 Figures produced (plots/)

Per thickness: overlay_log, overlay_linear, residual, band_diagram_{eq, vth, on} (E_c , E_v , E_{Fn} across the stack at the channel centre, depth origin at the film top surface), electron_density_{eq, vth, on} (across the film), electron_density_vs_vg (channel-centre and film-averaged free electrons; the $V_G = 0$ value is n_{FB}), mobility_vs_vg (native probe mobility vs the measured apparent field-effect mobility). Global: metrics_vs_thickness, thickness_laws_overview, mobility_roughness_factor.

10. Numerical verification (docs/NUMERICAL_CONVERGENCE.md)

Criteria: a check passes if the active-region log RMSE changes by < 0.01 dec, $|dV_{th}| < 10$ mV, $|dSS| < 3$ mV/dec, $|dIon| < 1$ %, and native zeros remain zeros.

check	evidence	result
A. lateral mesh x0.5 (2 nm, 28k nodes)	V0 run_0015 vs 0013	max 0.0046 dec, Ion -0.02 %: PASS
A. mesh x0.7 (13.2 nm, 20k nodes; x0.5 exceeds the 32-bit memory)	V0 run_0026 vs 0024	max 0.0040 dec: PASS
C. DOS levels 96/48 → 192/96	V0 run_0016 vs 0013	max 0.0056 dec, Ion +1.1 %: PASS (marginal on Ion)
E. interface layer 0.25 → 0.125 nm	V0 run_0017 vs 0013	max 0.009 dec: PASS
F. tolerance: CR.TOLER 1e-19 never converges; 1e-17 + XANDRNORM vs default	V0 runs 0006/0009/0010/0011	glitches removed, on-state unaffected: documented choice
G. Kirchhoff	Id+Is+Ig	at every bias
H. repeatability	V0 identical decks	identical currents: PASS
I. electrons-only vs two-carrier	Astra rebuild	identical Id; hole current 1e-38..1e-55 A: PASS
J. width convention	V0 WIDTH=2	raw currents x2.000, A/um identical: PASS
M. physical structure + stepped sweep vs reduced boundaries + per-point sweep	run_0012 vs 0008; run_0013 vs 0009	Id within 0.08 % at 0.5 V and 0.002 % at 3 V: PASS
A'. V1 final 13.2 nm deck, all spacings x0.7 (24k nodes)	run_0028 vs run_0013	max 0.0032 dec, Ion -0.01 %: PASS
C'. V1 final 2 nm deck, DOS levels 96/48 → 192/96 → 384/192	run_0019, run_0029 vs run_0012	Ion +2.0 % then +2.5 %, max 0.0125 dec, Vth/SS unchanged: FAILS the 1 % criterion; the on-current carries a +2.5 % systematic from the DOS discretization, smaller than the fit residual, so no conclusion changes; a recalibration at 384/192 would lower μ_{band} by ~2.5 %
K/L. 31.8 nm (excluded) and 6.3 nm mesh refinement	not run	OPEN

11. Comparison with the two earlier per-device calibrations

item	Astra 6 (2026-09-12)	Claude V0 (2026-09-10/21)	V1 (this)
band parameters	Eg 3.05, chi 4.30, eps 9.3, Nc 5e18 for all films	same	Eg(t), chi(t), m*(t), Nc(t) from laws
gate / contacts	WF 4.765; S/D WF 4.45 with surface recombination	WF 4.70; Ohmic	WF 4.70; Ohmic
flat band	one shared WF	per-device Qf	one shared Qf
tail	per-device NTA, WTA (0.035-0.040)	per-device NTA, WTA (0.036-0.044)	Nt(t) law, WTA 0.040 shared
Dit	none	per-device 2-6e11	3e11 shared
Nd	2e17 / 1e17 / 8.9e17	2e17 / 3e17 / 3e17 / 2.65e18	law 2.5e17 [1 + (t/20)^4]
mobility	13.3 / 12 / 55	12.3 / 11.3 / 55.1 / 82	18.1 / (12.4) / 61.9 band x roughness factor
active RMSE 2 / 6.3 / 13.2 nm	0.033 / 0.049 / 0.049	0.019 / 0.048 / 0.025	0.037 / 0.68 (validation) and 0.063 (tuned, one device-specific offset) / 0.081
31.8 nm	deferred	0.093 (back sheet + Rs)	excluded
tuned numbers	13 for three films	22+ for four films	1 shared + 1 per calibrated film

Robust, lineage-independent findings: the same effective tail capacity ($\sim 2e19$ cm⁻³ at 2 nm, falling $\sim 4\times$ by 13.2 nm), the same band-mobility step between 6.3 and 13.2 nm, and the same conclusion that intrinsic drift-diffusion cannot produce the off-state floors. V1 trades 0.03-0.06 dec of per-device accuracy for cross-thickness structure and exposes the 6.3 nm anomaly.

12. Limitations and things that were not done

1. No DFT of the target IWO: all confinement laws are pure-In₂O₃ PBE proxies; Quantum ESPRESSO could not be installed on this machine; inputs are prepared but unexecuted (qe_workflow/).
2. Absolute band alignment is not measured: chi_bulk, the TiN work function and Qf are degenerate; only thickness differences of chi are used as physics. C-V or Kelvin probe per thickness would remove the degeneracy.
3. The composition "2 % W" is undefined and never converted to a donor density.
4. Off-state floors are not modelled; their origin is NOT DETERMINED (no IG/IS or instrument-floor records).
5. Mobility: constant band mobility per film times the roughness factor; the gate roll-off of Eq. (6) is not insertable natively; μ_{band} is fitted and the 6.3→13.2 nm step has no law.

6. The 6.3 nm validation fails on the threshold by 0.29 V (Section 9).
7. The 31.8 nm device is excluded: laws + series resistance give I_{on} within 6 % but not the flat always-on baseline; the back-surface donor-sheet hypothesis run aborted in SOLVE INIT and was not investigated.
8. D_{it} is an assumed shared value; capture cross sections, lifetimes, N_v and the deep Gaussian are assumed and not identifiable from DC curves.
9. Quasi-2D sub-band quantisation of the in-plane DOS is not represented (Approach A uses a 3-D continuum N_c with a confinement-corrected mass).
10. Calibration, not validation, for 2 and 13.2 nm: one transfer curve per device at one VD and one temperature; no ID-VD, temperature or C-V data were available.
11. Contacts are ideal; the 2 μm overlap is a numerical assumption.
12. The no-confinement counterfactual, eight one-at-a-time perturbations at 2 nm and the V1 mesh/DOS refinements were executed on 2026-09-22 (Sections 8.1 and 10). Not run: gate work function (same action as Qf), m^* alone, sensitivity cases at 6.3 nm, and the mesh refinement of the 6.3 nm film.
13. run_0010 and run_0011 produced no currents (aborted / stopped) and are recorded as such.
14. Data provenance: current units, VD and geometry come from the schematic, not from workbook headers.

Integrity statement: no material parameter or source was invented; no fit is called a measurement; the W content was not treated as a donor density; pure-In₂O₃ values are labelled as proxies; no PBE gap is used as an experimental gap; the fitted Qf is declared as an effective alignment correction and was shown to be a common offset rather than a thickness-dependent patch; mobility was not used to correct a wrong carrier density (it was fitted last, at fixed electrostatics, and moved V_{th} by < 10 mV); no ATLAS run, DFT run or convergence test is fabricated; no nonzero leakage was manufactured where ATLAS gives zero; ATHENA was not used; "exact match" is not claimed anywhere; the model is not called predictive.

13. Publication-readiness statement

The package is publication-ready as a calibrated, thickness-resolved TCAD parameter extraction for the 2, 6.3 and 13.2 nm devices (0.037 / 0.063 / 0.081 dec) with one documented shared electrostatic offset, one documented device-specific offset for the 6.3 nm film, and a documented negative out-of-sample result for that film under the shared laws. It is not a predictive model of thickness scaling: the shared laws miss the 6.3 nm threshold by 0.29 V, the band mobility is fitted per film, the confinement law is a pure-In₂O₃ proxy, and no independent bias, temperature or capacitance data were available. A manuscript should describe it as calibrated on the same transfer curves it reproduces and report both the 6.3 nm validation miss and its device-specific correction. A ready-to-edit Methods paragraph is in docs/PUBLICATION_METHODS_DRAFT.md.

What would make it predictive: C-V (or Kelvin probe) per thickness; a second bias or temperature per device; DFT of W-doped slabs; TLM for the contacts; gate/source current records for the off state.

14. How to reproduce

From the package directory with the Python environment of the sibling package (`..\IWO_ATLAS_Model_claude\.venv\Scripts\python.exe`):

```
python scripts/material_laws.py                # laws -> config/iwo_material_model.yaml,
tables/THICKNESS_LAWS.csv, law plots
python scripts/build_iwo_decks.py              # decks/iwo_{2p0,6p3,13p2,31p8}nm.in in the final form
(physical structure, stepped sweep)
python scripts/run_atlas.py --thickness 2 --label rerun --override
traps.fixed_interface_charge_Qf_cm2.value=1.73e12 --override
"thickness_laws.mu0_cm2Vs.mu_band_fitted[2.0]=18.13"
```

```
python scripts/run_atlas.py --thickness 13.2 --label rerun --override
traps.fixed_interface_charge_Qf_cm2.value=1.73e12 --override
"thickness_laws.mu0_cm2Vs.mu_band_fitted[13.2]=61.9"
python scripts/run_atlas.py --thickness 6.3 --label validation --override
traps.fixed_interface_charge_Qf_cm2.value=1.73e12
python scripts/compare_experiment_simulation.py      # tables/EXPERIMENTAL_VS_SIMULATION.{csv,md} from
results/BEST_RUNS_V1.json
python scripts/plot_results.py                      # plots/
python scripts/make_provenance_tables.py            # tables/MATERIAL_PARAMETERS_USED.*,
PARAMETER_PROVENANCE.csv
python scripts/sensitivity_from_stage_runs.py        # tables/SENSITIVITY_RESULTS.*
```

Every launch creates a new `results/runs/run_NNNN_*` directory (deck, DeckBuild log, transfer.log, exports, KCL table, comparison, execution record) and appends to `results/RUN_INDEX.csv`; the launch budget is enforced by `config/solver.yaml`.

15. References

Supplied and read (PDF text in `inputs/papers_text/`):

1. M. Januar, Z.-F. Luo, K.-C. Liu, M.-H. Lee, "Unified Analytic Framework for Thickness-Dependent Transport and Trap-State Modulation in Ultrathin W:In₂O₃ Field-Effect Transistors," *Small Structures* 7, e202500807 (2026), DOI 10.1002/sstr.202500807. Stack, process, Eqs. (1)-(9), Nt/Tt, Δsr, mobility law, DFT trends. Supporting Information not supplied (eps values recorded from the online SI by the Astra audit).
2. M. Si, Y. Hu, Z. Lin, et al., P. D. Ye, "Why In₂O₃ Can Make 0.7 nm Atomic Layer Thin Transistors," *Nano Letters* 21, 500 (2021), DOI 10.1021/acs.nanolett.0c03967. CNL 0.4 eV above Ec; DFT dEc ≈ 0.6 eV at 1.5 nm with Ev unchanged.
3. Z. Lin, M. Si, V. Askarpour, et al., P. D. Ye, "Nanometer-Thick Oxide Semiconductor Transistor with Ultra-High Drain Current," *ACS Nano* 16, 21536 (2022), DOI 10.1021/acsnano.2c10383. PBE gaps 0.94/1.27/1.88 eV; m* 0.17/0.19/0.23/0.30 m0; Dit 6e11.
4. Z. Wang, Z. Lin, M. Si, P. D. Ye, "Characterization of Interface and Bulk Traps in Ultrathin Atomic Layer-Deposited Oxide Semiconductor MOS Capacitors With HfO₂/In₂O₃ Gate Stack by C-V and Conductance Method," *Frontiers in Materials* 9, 850451 (2022), DOI 10.3389/fmats.2022.850451. Dit 6.3e11; bulk DOS 3.3e20.
5. M. Stokely, R. Korlacki, S. Knight, et al., M. Schubert, "Optical phonon modes, static and high-frequency dielectric constants, and effective electron mass parameter in cubic In₂O₃," *J. Appl. Phys.* 129, 225102 (2021), DOI 10.1063/5.0052848. eps_DC 10.55, m* 0.208 m0.
6. H. Kim, H.-S. Choi, G. Yun, W.-J. Cho, H. Park, "Understanding thickness-dependent stability of tungsten-doped indium oxide transistors," *Appl. Phys. Lett.* 125, 173507 (2024), DOI 10.1063/5.0228363. Thickness trend of bulk/border traps.
7. M. Si, Z. Lin, Z. Chen, X. Sun, H. Wang, P. D. Ye, "Scaled indium oxide transistors fabricated using atomic layer deposition," *Nature Electronics* 5, 164 (2022), DOI 10.1038/s41928-022-00718-w. Contact/CNL context.
8. N. Pandey et al., "Analytical Modeling of Short-Channel Effects in BEOL-Compatible Thin-Film Transistors," *IEEE Trans. Electron Devices* 72, 2381 (2025), DOI 10.1109/TED.2025.3546593. TCAD trap convention (context).
9. X. Wang et al., "A Physics-Based Compact Model for IGZO Channel FET Toward Subthreshold Characteristic Dependent Memory Application," *IEEE Trans. Electron Devices* 72, 2390 (2025), DOI 10.1109/TED.2025.3549745. Donor-Gaussian treatment (context).
10. K. Anusha, A. D. D. Dwivedi, "Review and analysis on numerical simulation and compact modeling of InGaZnO thin-film transistor for display SENSOR applications," *Measurement: Sensors* 36, 101391 (2024), DOI 10.1016/j.measen.2024.101391. Background.

11. P. Giannozzi et al., "QUANTUM ESPRESSO: a modular and open-source software project for quantum simulations of materials," J. Phys.: Condens. Matter 21, 395502 (2009), DOI 10.1088/0953-8984/21/39/395502. Software reference for the unexecuted workflow.

Cited inside the packages and used as priors (not supplied; METADATA_REQUIRES_VERIFICATION):

12. W.-T. Fan et al., "Numerical Analysis of Oxygen-Related Defects in Amorphous In-W-O Nanosheet Thin-Film Transistor," Nanomaterials 11, 3070 (2021), DOI 10.3390/nano11113070. Eg 3.05 / chi 4.30 priors, deep-acceptor range.
13. C. Yoo et al., "Atomic Layer Deposition of WO₃-Doped In₂O₃ for Reliable and Scalable BEOL-Compatible Transistors," Nano Letters 24, 5737 (2024), DOI 10.1021/acs.nanolett.4c00746. W content evidence.
14. I. Hamberg, C. G. Granqvist, J. Appl. Phys. 60, R123 (1986). eps 8.9 of evaporated films.
15. P. D. C. King et al., Phys. Rev. Lett. 101, 116808 (2008). CNL above Ec.
16. J. Robertson, S. J. Clark, Phys. Rev. B 83, 075205 (2011). CNL.
17. Y. Hinuma, T. Gake, F. Oba, Phys. Rev. Materials 3, 084605 (2019). In₂O₃/Al₂O₃ band alignment.
18. M. Feneberg et al., In₂O₃ nonparabolicity, m*(0) = 0.18 m₀ (cited in ref. 5).
19. T. Ando, A. B. Fowler, F. Stern, Rev. Mod. Phys. 54, 437 (1982). Surface-roughness mobility law.
20. J. Lee, K.-H. Lim, Y. S. Kim, Sci. Rep. 8, 13905 (2018). Gate-leakage caution.

Software and manuals: Silvaco ATLAS User's Manual 5.28.1.R (DEFECTS, INTDEFECTS, CONTACT, METHOD, TFT and quantum chapters, installed copy); Silvaco DeckBuild User's Manual 5.0.10.R (batch options); installed vendor TFT examples used to validate syntax. Packages audited: GPT Astra 6 IWO_PRIORITY_THREE_20260912.zip; Claude IWO_ATLAS_Model_claude.

Appendix A. Launch index (results/RUN_INDEX.csv)

run	t (nm)	label	outcome
0001	2.0	laws only	scored, 3.93 dec (offset +0.33 V)
0002	13.2	laws only	scored, 1.03 dec (offset +0.29 V)
0003	2.0	+ shared Qf 1.73e12	0.056 dec
0004	13.2	+ shared Qf	0.086 dec
0005	6.3	shared parameters (prediction)	0.678 dec, Vth -0.29 V
0006	31.8	laws + shared Qf + Rs	1.05 dec; baseline not reproduced
0007	6.3	device-specific Qf 8.7e10 (evidence)	0.082 dec
0008	2.0	μ _{band} 18.13	0.037 dec
0009	13.2	μ _{band} 61.9	0.081 dec
0010	31.8	back-surface sheet hypothesis, stepped	aborted in SOLVE INIT (no error text)

run	t (nm)	label	outcome
0011	6.3	μ _band 11.69	stopped by the user before completion
0012	2.0	FINAL: physical structure, stepped sweep	0.037 dec (rescored under the median-floor KCL rule)
0013	13.2	FINAL	0.081 dec
0014	6.3	FINAL validation, shared parameters	0.678 dec
0015	6.3	FINAL tuned: device-specific Qf 8.7e10 + μ _band 11.69	0.063 dec
0016 / 0017	2.0 / 13.2	no-confinement counterfactual	dVth -0.316 V / -0.023 V
0018	13.2	mesh x0.7 (900 s timeout)	timed out at 68/76 points; repeated as 0028
0019 / 0029	2.0	DOS levels 192/96 / 384/192	Ion +2.0 % / +2.5 %
0020-0027	2.0	one-at-a-time: chi +/-0.1, Nd x2, Nt x1.5, WTA +5 meV, Dit x3, eps 10.55, overlap 4 um	Section 8.1
0028	13.2	mesh x0.7 (1500 s timeout)	max 0.0032 dec: PASS
0016 / 0017	2.0 / 13.2	no-confinement counterfactual	Vth -0.316 V / -0.023 V
0018	13.2	mesh x0.7	timed out at 900 s (68 of 80 points)
0019 / 0029	2.0	DOS levels 192/96 / 384/192	Ion +2.0 % / +2.5 %
0020-0027	2.0	one-at-a-time: chi +/-0.1, Nd x2, Nt x1.5, WTA +5 meV, Dit x3, eps 10.55, overlap 4 um	Section 8.1
0028	13.2	mesh x0.7, 1500 s	max 0.0032 dec: PASS

Appendix B. Metric definitions (scripts/extract_metrics.py, applied identically to both curves)

Vth_cc: gate voltage where $I_d = 1e-9$ A/um, log-linear interpolation between adjacent points, no extrapolation. Vth_lin: linear extrapolation at maximum transconductance. SS_min: minimum over 0.2 V centred windows of $dV_G/d\log_{10}(I_d)$, restricted to currents above five times the MEASURED off-band median of the same device (applied to the simulated curve too, so both are evaluated over the same current range) and below 1 % of Ion. SS_cc: average slope between the $1e-10$ and $1e-8$ A/um crossings. $\mu_{FE} = gm_{max} L / (W Cox VD)$ with $L = 20$ um, $W = 1$ um for A/um data, $Cox = 8.955e-7$ F/cm², VD

= 0.7 V (an apparent field-effect mobility, not the band mobility). $I_{on} = I_d(+3 \text{ V})$. I_{off} = minimum I_d (measurement) or minimum POSITIVE I_d with the count of native zeros (simulation). Threshold differences are absolute; no percentage for sign-changing quantities. Active-region log RMSE: RMS of $\log_{10}(I_{sim}/I_{meas})$ over measured points above five times the off-band median.

Appendix C. Glossary of ATLAS keywords used

MESH WIDTH: third-dimension width (um). REGION/ELECTRODE: geometry; `user.material` = user-defined material. DOPING UNIFORM N.TYPE CONC: ionised donor density (cm⁻³). MATERIAL EG300 / AFFINITY / PERMITTIVITY / NC300 / NV300 / MUN / MUP / TAUN0 / TAUP0: band gap (eV), electron affinity (eV), relative permittivity, effective DOS (cm⁻³), electron/hole mobility (cm²/Vs), SRH lifetimes (s). CONTACT WORKFUNCTION: metal work function (eV); CONTACT RESISTANCE: lumped resistance (ohm·um in 2-D). MODELS FERMI SRH: Fermi statistics, SRH recombination. DEFECTS CONTINUOUS NUMA/NUMD: continuous sub-gap DOS with the given number of acceptor/donor energy levels; NTA/NTD: tail intercepts at E_c/E_v (cm⁻³ eV⁻¹); WTA/WTD: tail widths (eV); NGA/NGD: Gaussian amplitudes (cm⁻³ eV⁻¹); EGA (from E_c)/EGD (from E_v): Gaussian centres (eV); WGA/WGD: Gaussian widths; SIGTAE.. SIGGDH: capture cross sections (cm²); AFILE/DFILE: DOS export files. INTERFACE QF: fixed interface charge (cm⁻²). METHOD NEWTON TRAP MAXTRAPS ITLIMIT CLIMIT IR.TOL CR.TOLER XANDRNORM CARRIERS: Newton solver, bias-step halving, iteration limit, concentration limit, absolute current tolerance, continuity residual tolerance, dual-norm acceptance, carrier selection. PROBE: named scalar recorded at each bias. SOLVE INIT / VSTEP / VFINAL / NAME: equilibrium solution; stepped bias sweep on the named electrode. LOG OUTF: logged sweep file. SAVE OUTF: structure file. EXTRACT: DeckBuild post-processing (curves from logs and cut-lines from structures).

Appendix D. Figures

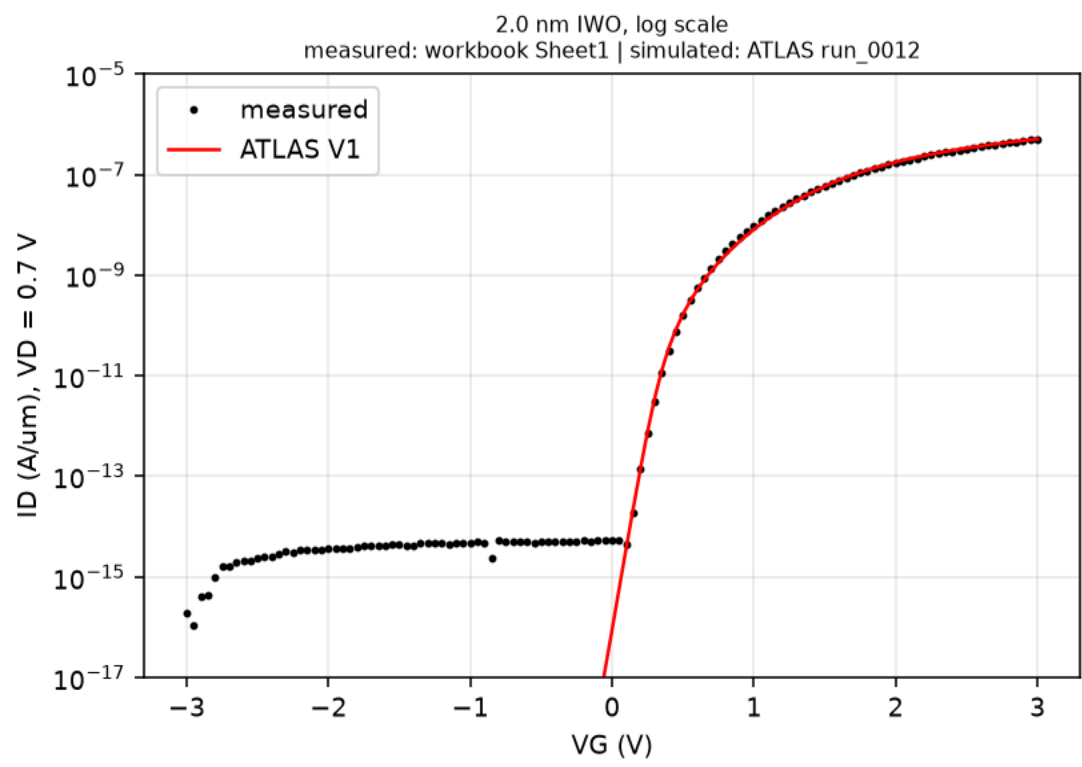


Figure F1. 2.0 nm: measured (workbook) vs ATLAS run_0012, log scale. (plots/2p0/overlay_log.png)

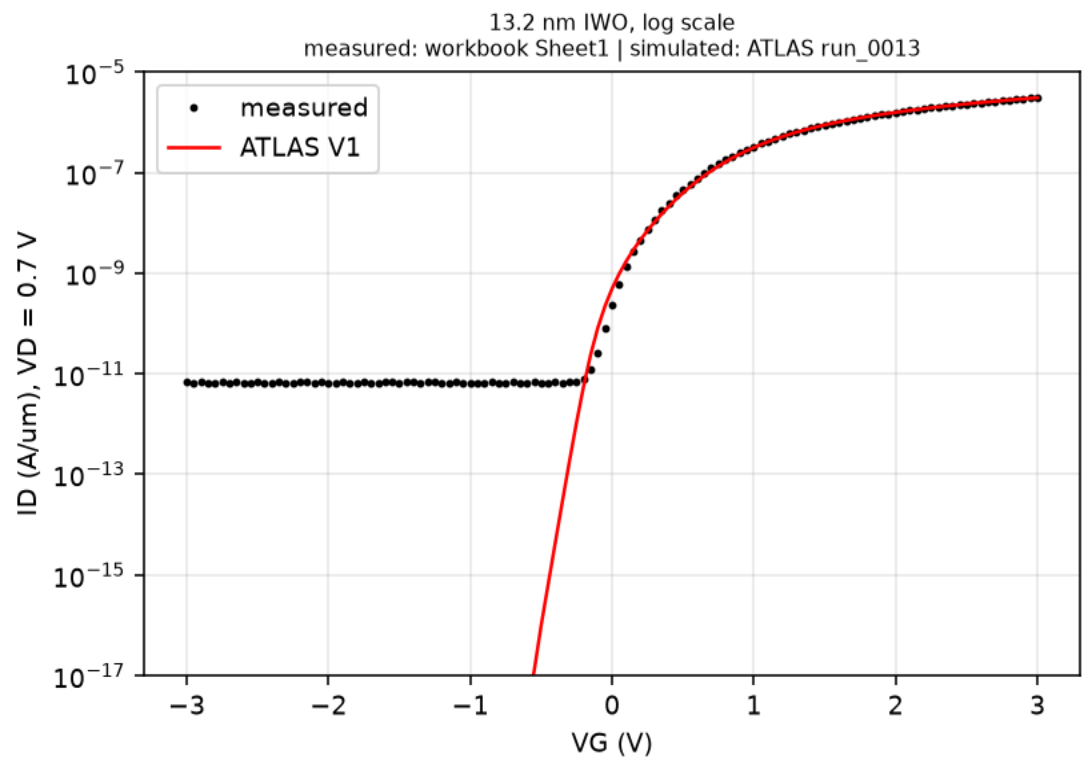


Figure F2. 13.2 nm: measured vs ATLAS run_0013, log scale. (plots/13p2/overlay_log.png)

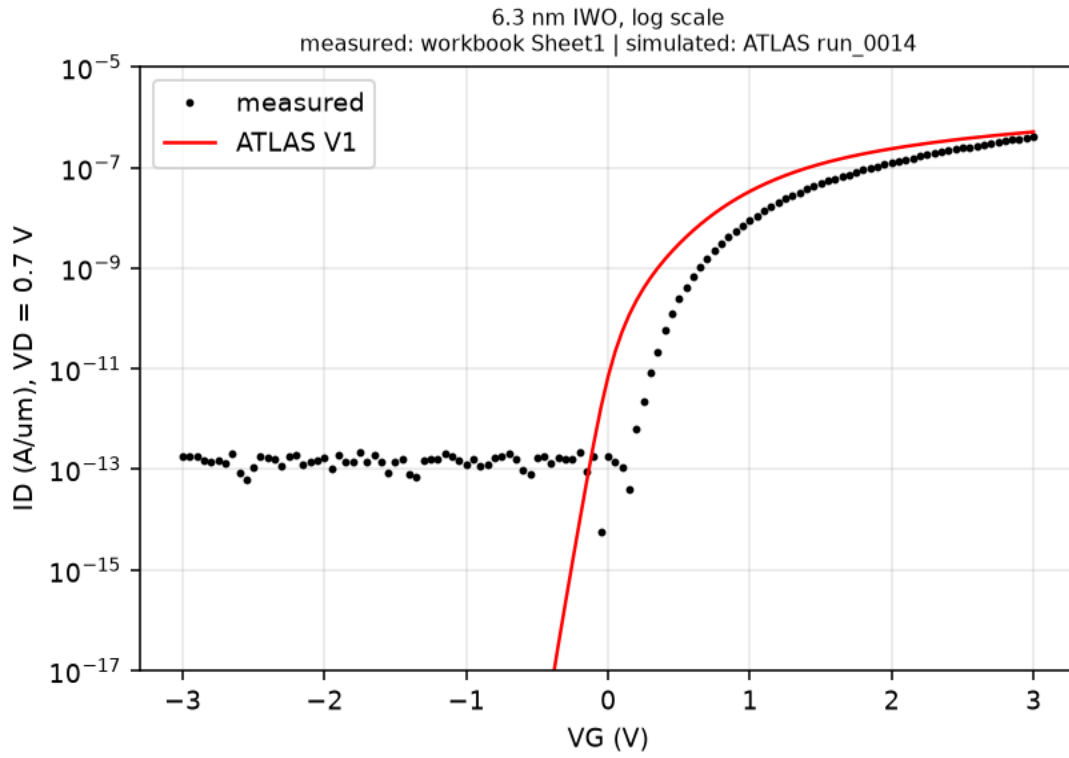


Figure F3. 6.3 nm validation with the shared parameters: measured vs ATLAS run_0014 (threshold miss of 0.29 V).
(plots/6p3_validation/overlay_log.png)

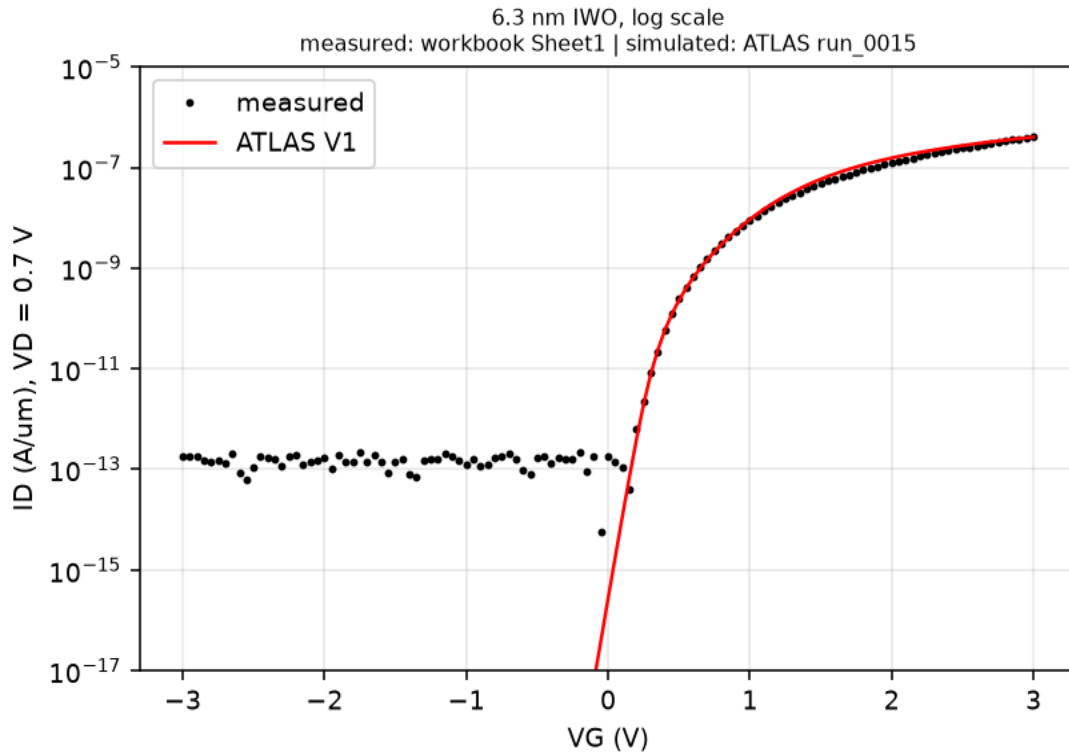


Figure F3b. 6.3 nm tuned (device-specific flat-band offset + band mobility): measured vs ATLAS run_0015. (plots/6p3/overlay_log.png)

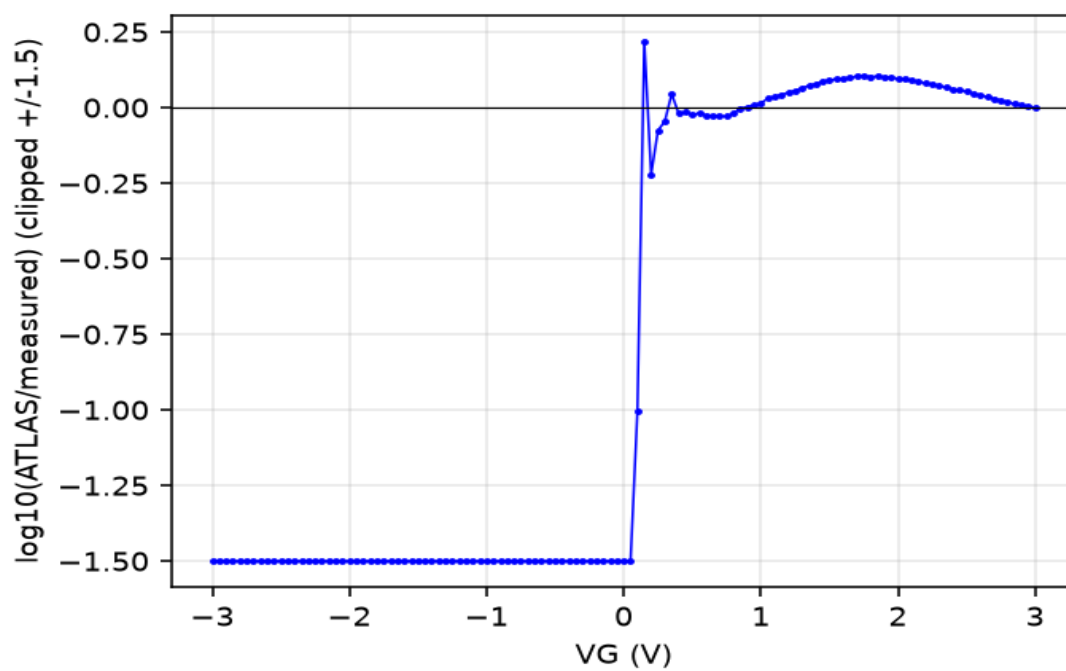


Figure F3c. 6.3 nm tuned residuals (run_0015). (plots/6p3/residual.png)

2.0 nm residuals (run_0012); native zeros where log residual is clipped

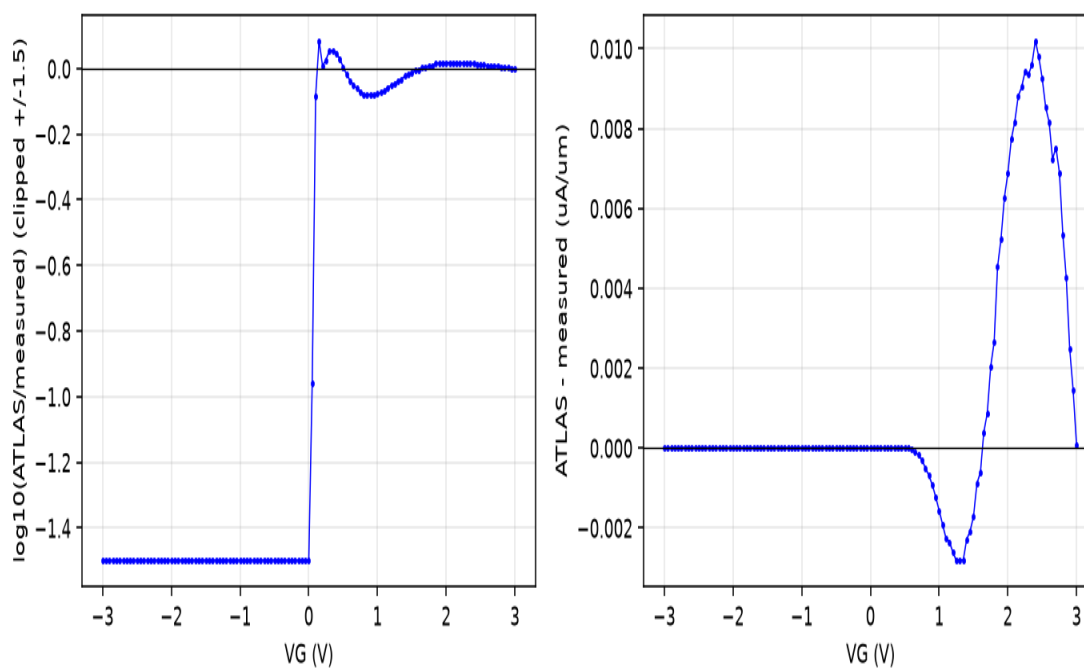


Figure F4. 2.0 nm residuals (log and linear). (plots/2p0/residual.png)

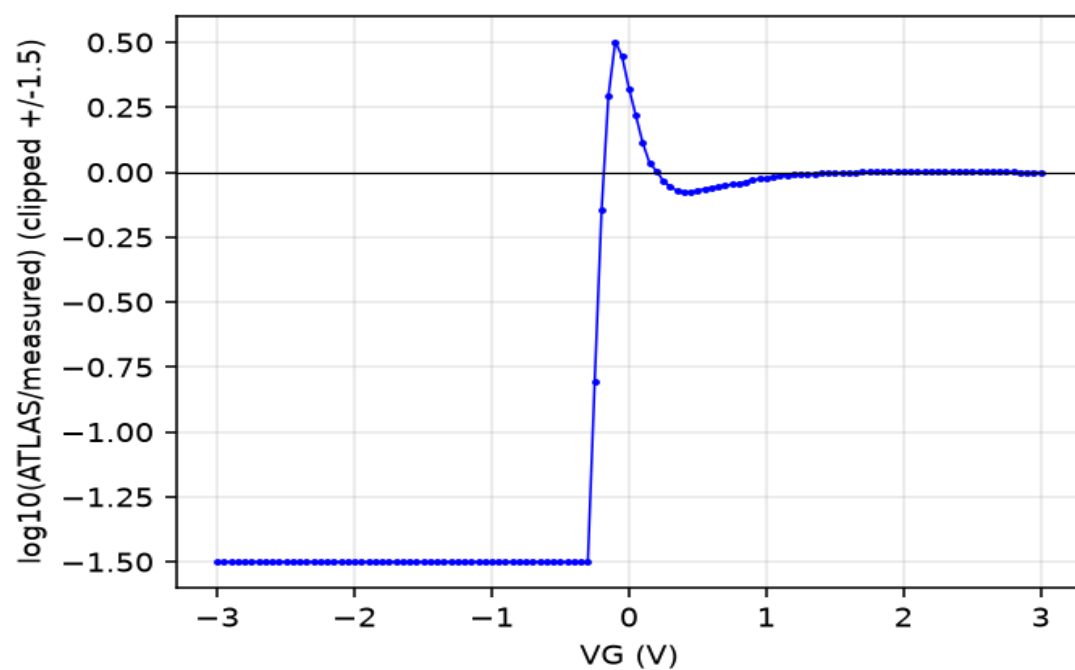


Figure F5. 13.2 nm residuals. (plots/13p2/residual.png)

Device metrics vs thickness: experiment (workbook) vs ATLAS V1 best runs, identical extraction

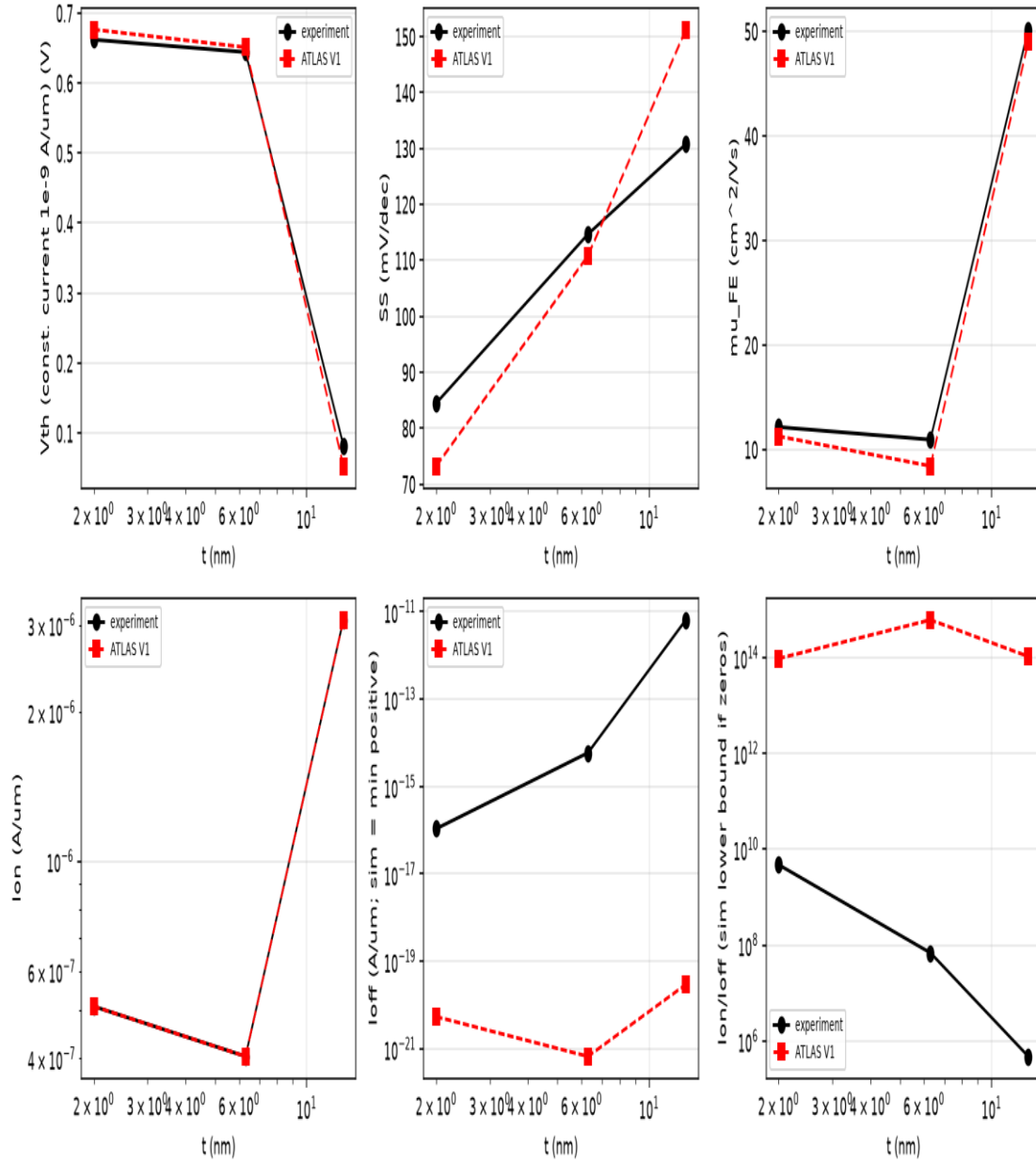


Figure F6. Device metrics vs thickness, experiment vs ATLAS, identical extraction. (plots/metrics_vs_thickness.png)

IWO_material(t) V1 thickness laws: DFT proxies (pure In2O3), measured bulk m^* , fitted Nt/Nd laws

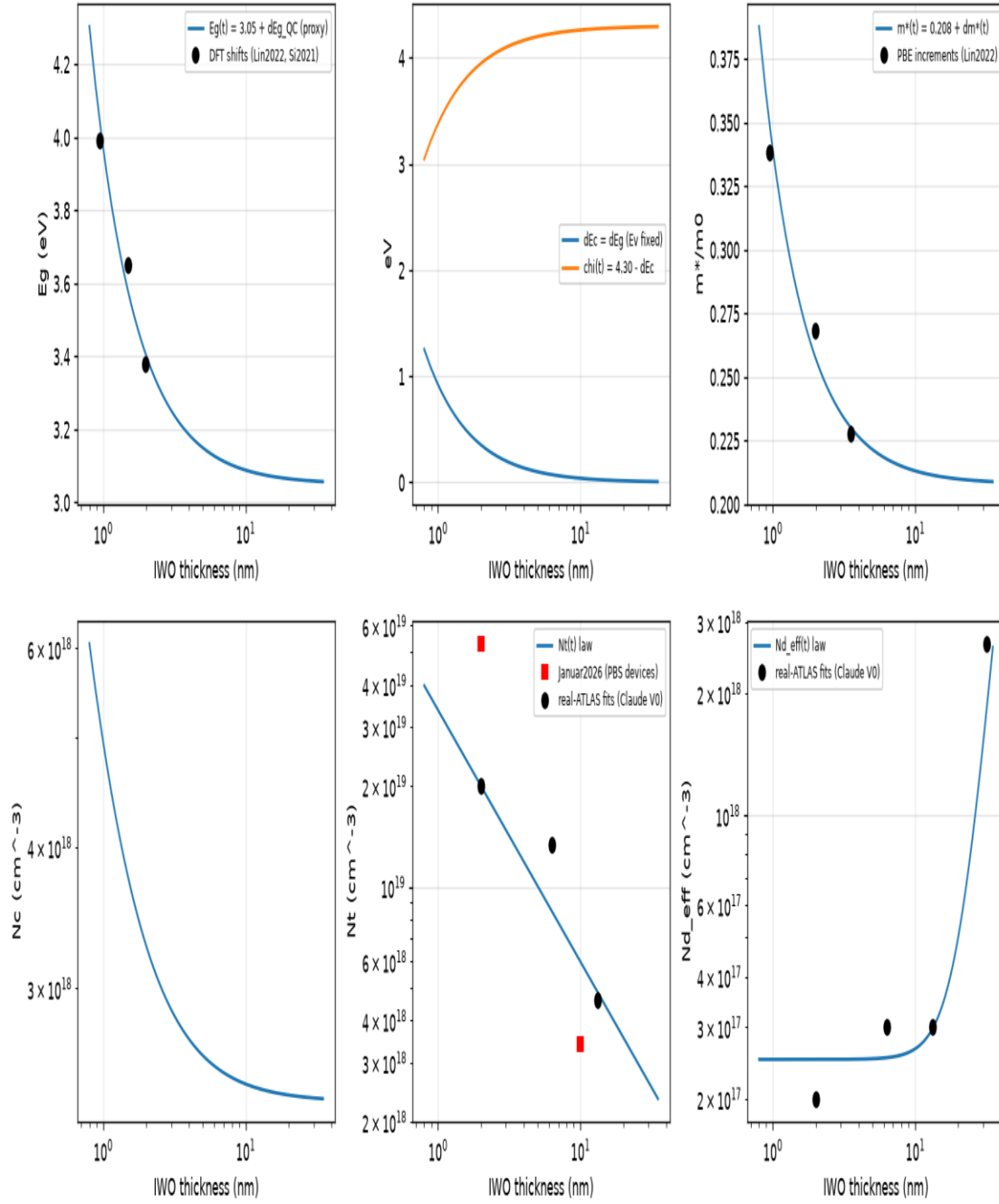
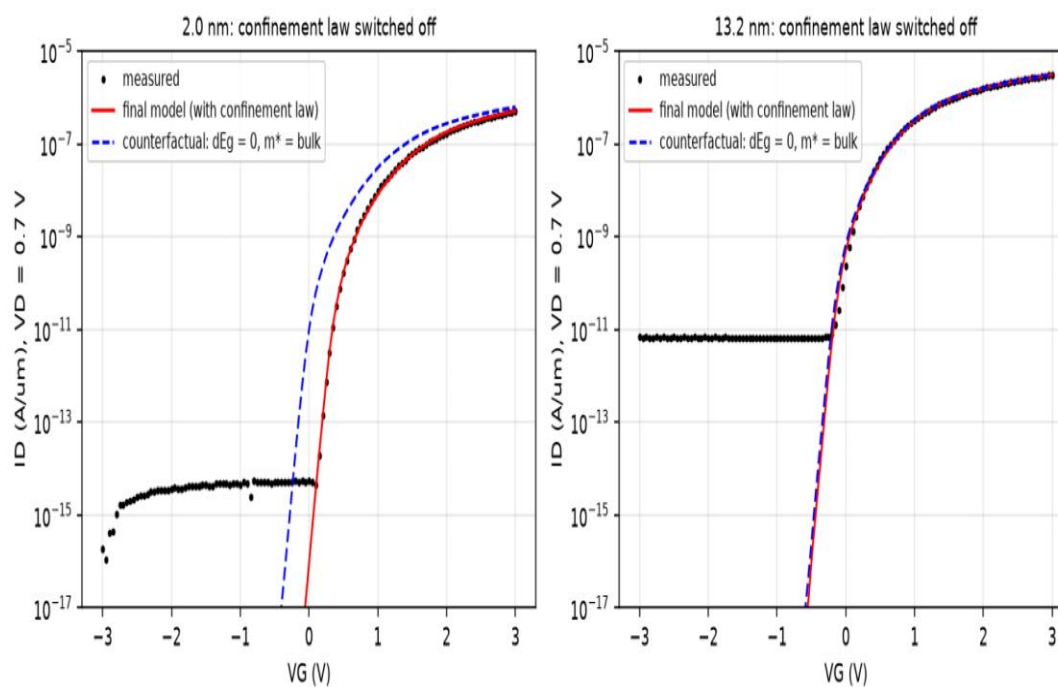


Figure F7. The thickness laws of the material model with their evidence points. (plots/thickness_laws_overview.png)

No-confinement counterfactual (runs 0016/0017 vs 0012/0013): 2 nm shifts by -0.32 V, 13.2 nm by -0.02 V



One-at-a-time sensitivity at 2 nm vs the final run_0012 (runs 0020-0027)

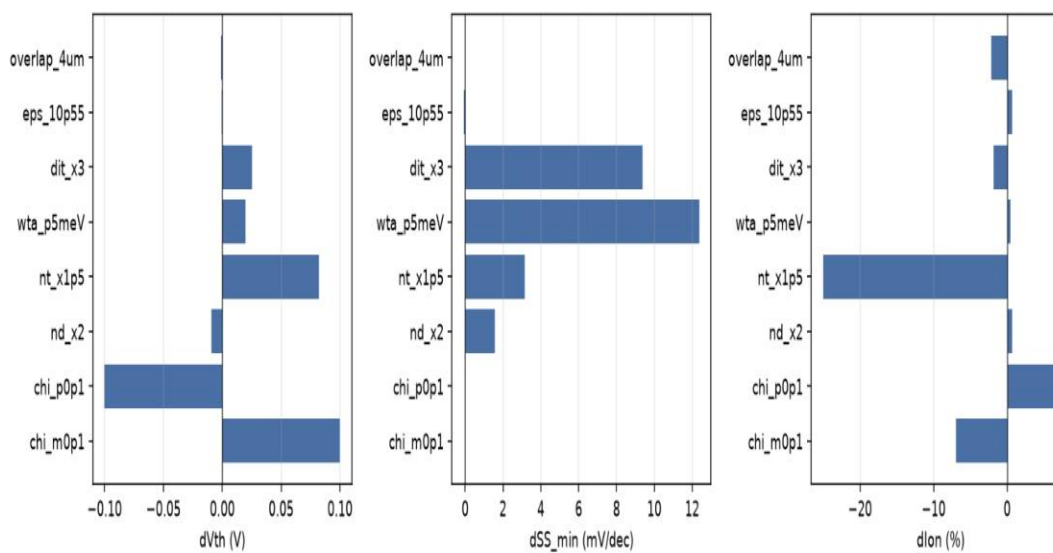


Figure F7b. No-confinement counterfactual overlays (top) and one-at-a-time sensitivities at 2 nm (bottom), runs 0016-0027. (plots/sensitivity_overview.png)

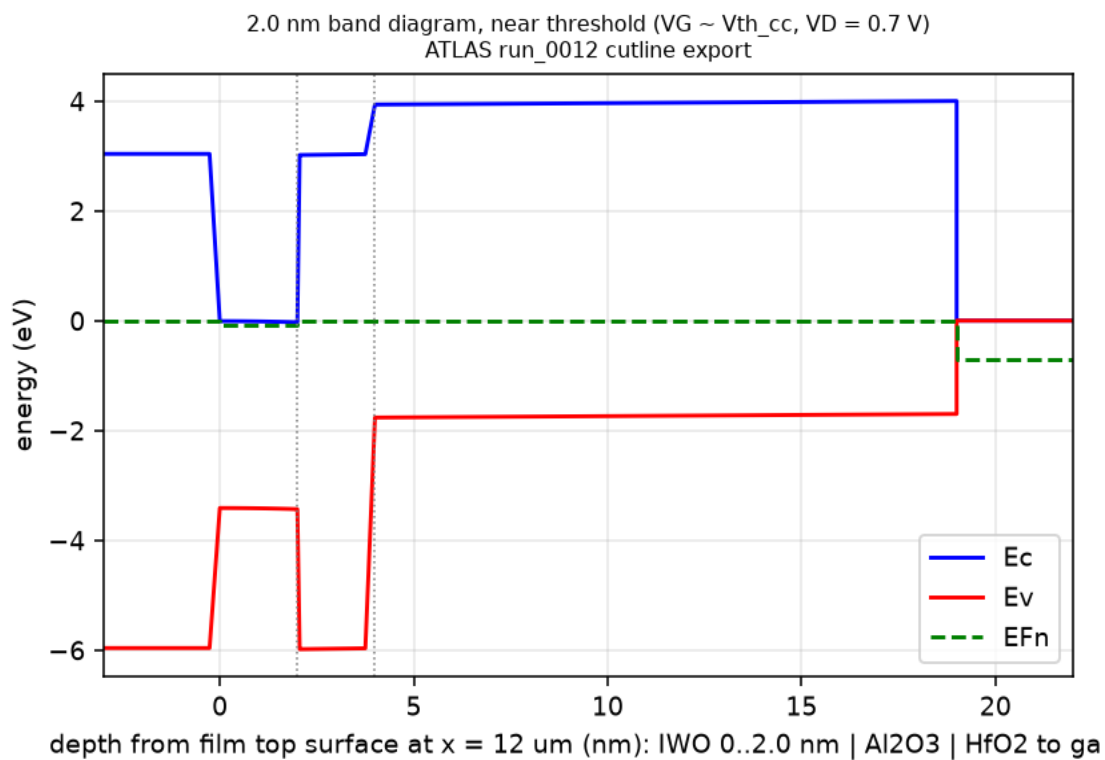


Figure F8. 2.0 nm band diagram near threshold across the stack (channel centre). (plots/2p0/band_diagram_vth.png)

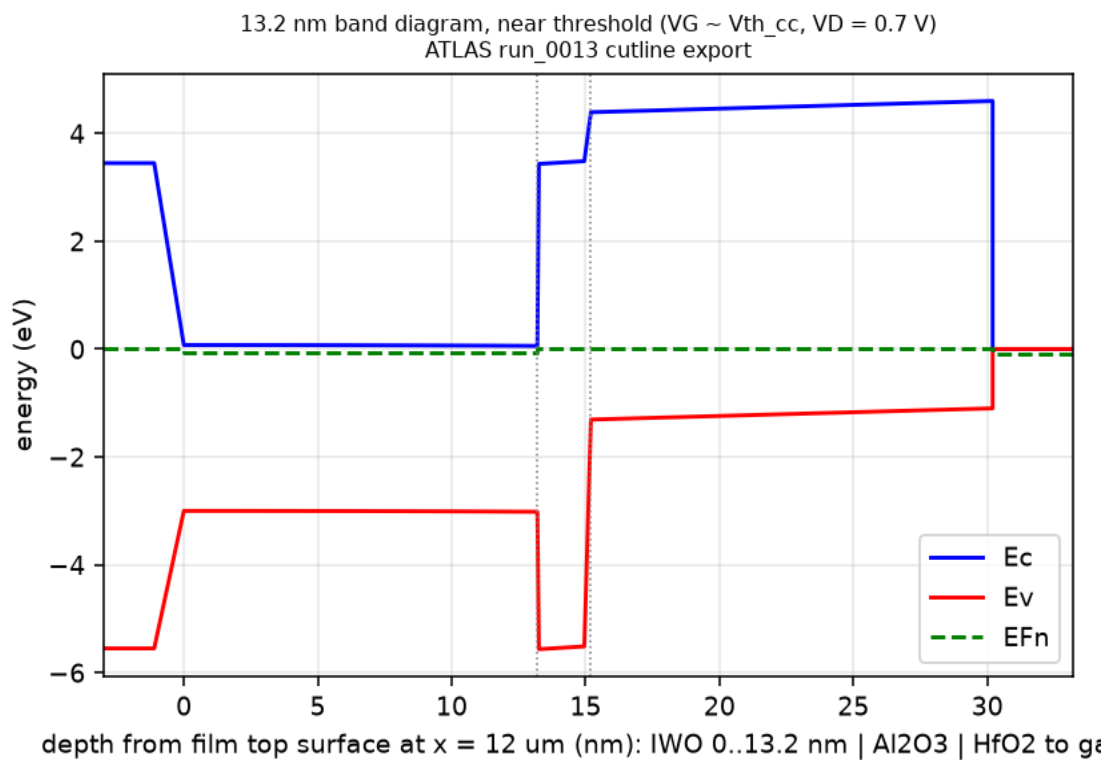


Figure F9. 13.2 nm band diagram near threshold. (plots/13p2/band_diagram_vth.png)

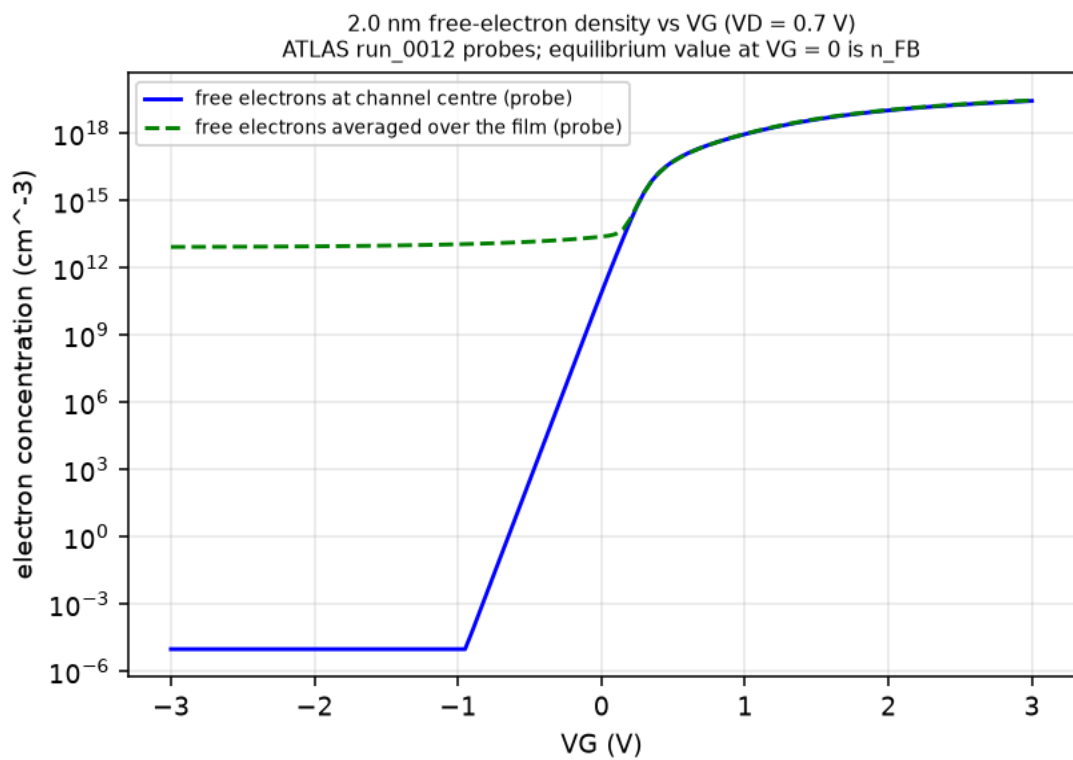


Figure F10. 2.0 nm free-electron density vs VG (channel centre and film average; VG = 0 value is n_{FB}).
(plots/2p0/electron_density_vs_vg.png)

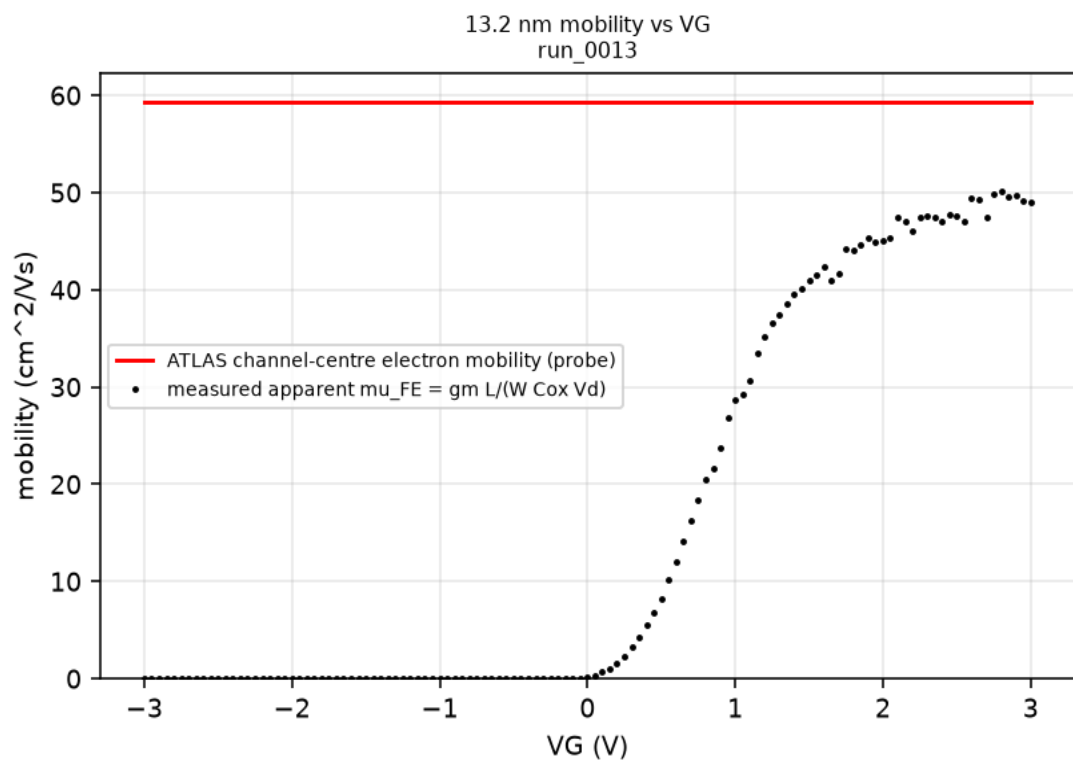


Figure F11. 13.2 nm: native probe mobility vs the measured apparent field-effect mobility. (plots/13p2/mobility_vs_vg.png)